US 20250274979A1

(19) United States
(12) Patent Application Publication
Bansal et al.
(10) Pub. No.: US 2025/0274979 A1
(43) Pub. Date: Aug. 28, 2025

(54) RTS-CTS PROCEDURE FOR EXTENDED RANGE PACKET TRANSMISSION

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(21) Appl. No.: **19/006,694**

(22) Filed: **Dec. 31, 2024**

(30) **Foreign Application Priority Data**

Feb. 28, 2024 (IN) 202441014419
Feb. 28, 2024 (IN) 202441014478

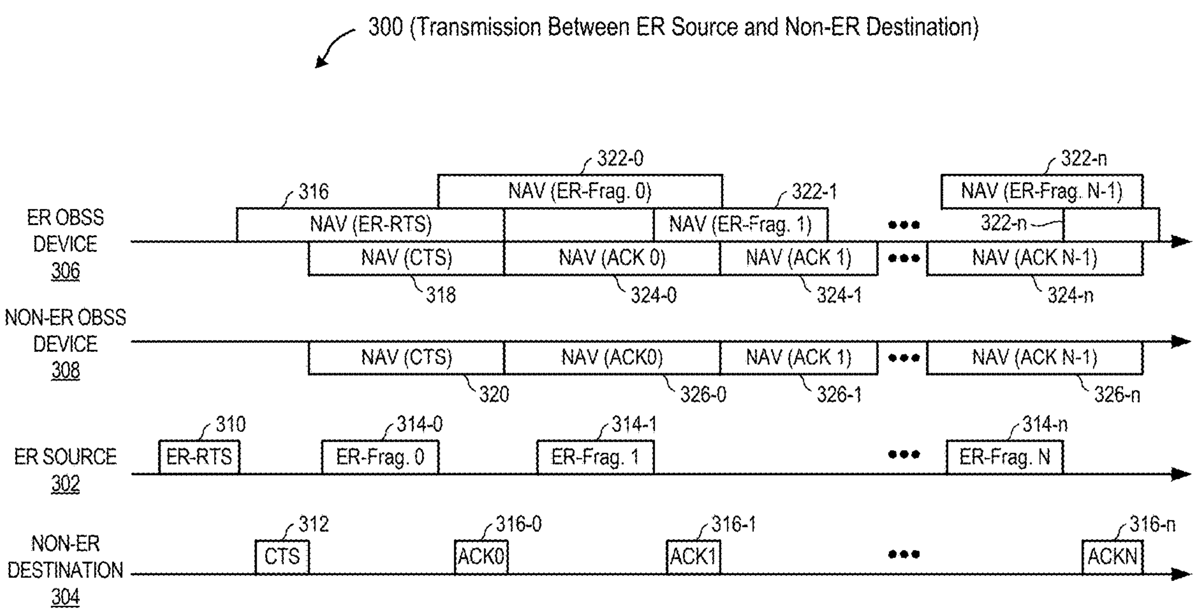
Publication Classification

(51) **Int. Cl.**
H04W 74/0816 (2024.01)

(52) **U.S. Cl.**
CPC *H04W 74/0816* (2013.01)

(57) **ABSTRACT**

Methods and apparatus for wireless channel reservation using extended range Request to Send (RTS)/Clear to Send (CTS) procedures to minimize collisions with OBSS devices. In a method, a first wireless device transmits an extended range RTS frame and receives, in response, a non-extended range CTS frame from a second wireless device. The first wireless device then transmits an extended range physical layer protocol data unit (PPDU) for reception by the second wireless device. This method may further include fragmenting the extended range PPDU into a plurality of extended range fragments, transmitting a first extended range fragment, receiving a non-extended range acknowledgement (ACK) from the second wireless device, and transmitting a second extended range fragment.



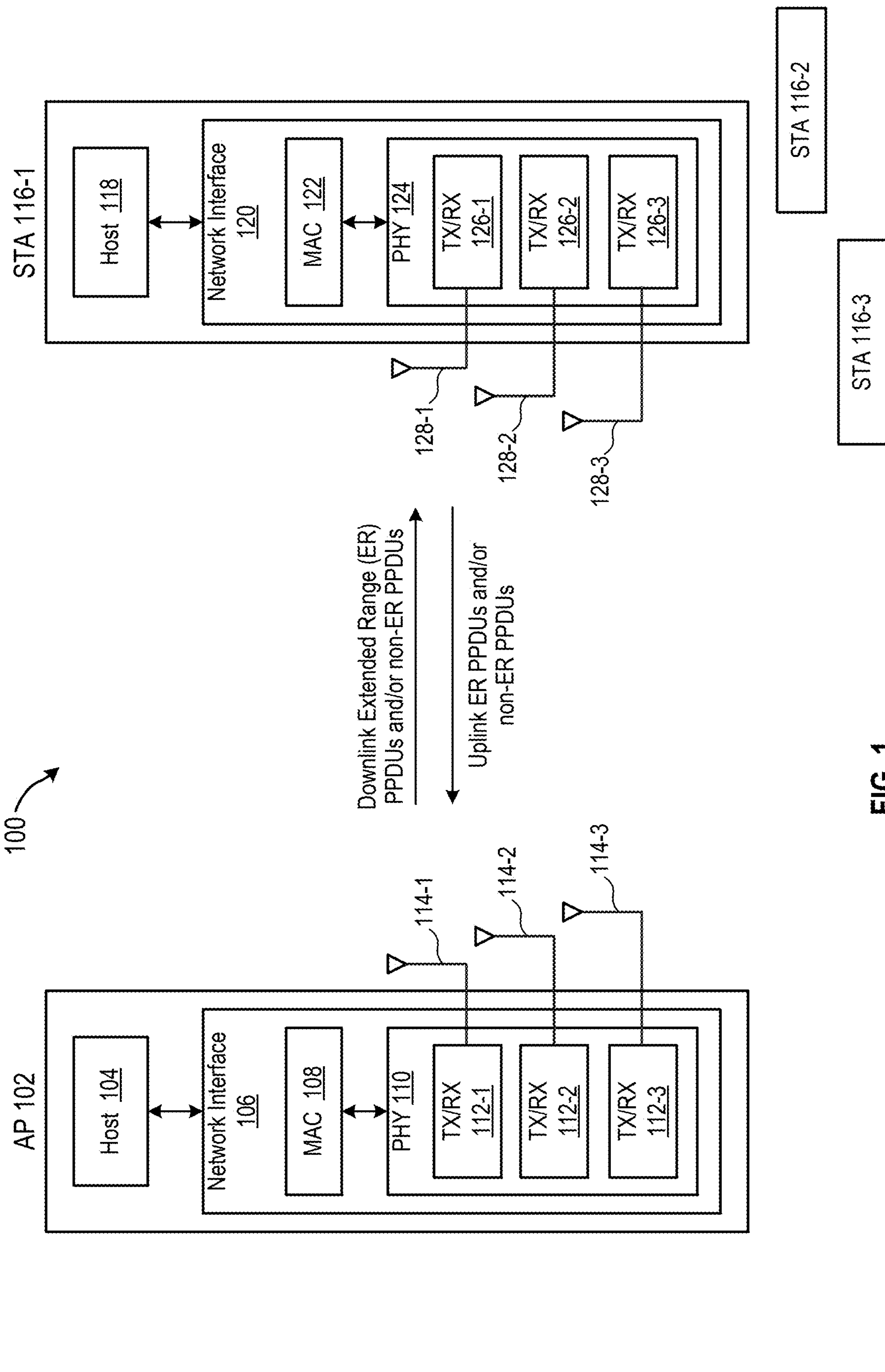

100
AP 102
Host 104
Network Interface 106
MAC 108
PHY 110
TX/RX 112-1
TX/RX 112-2
TX/RX 112-3
114-1
114-2
114-3
Downlink Extended Range (ER) PPDUs and/or non-ER PPDUs
Uplink ER PPDUs and/or non-ER PPDUs
STA 116-1
Host 118
Network Interface 120
MAC 122
PHY 124
TX/RX 126-1
TX/RX 126-2
TX/RX 126-3
128-1
128-2
128-3
STA 116-2
STA 116-3

FIG. 1

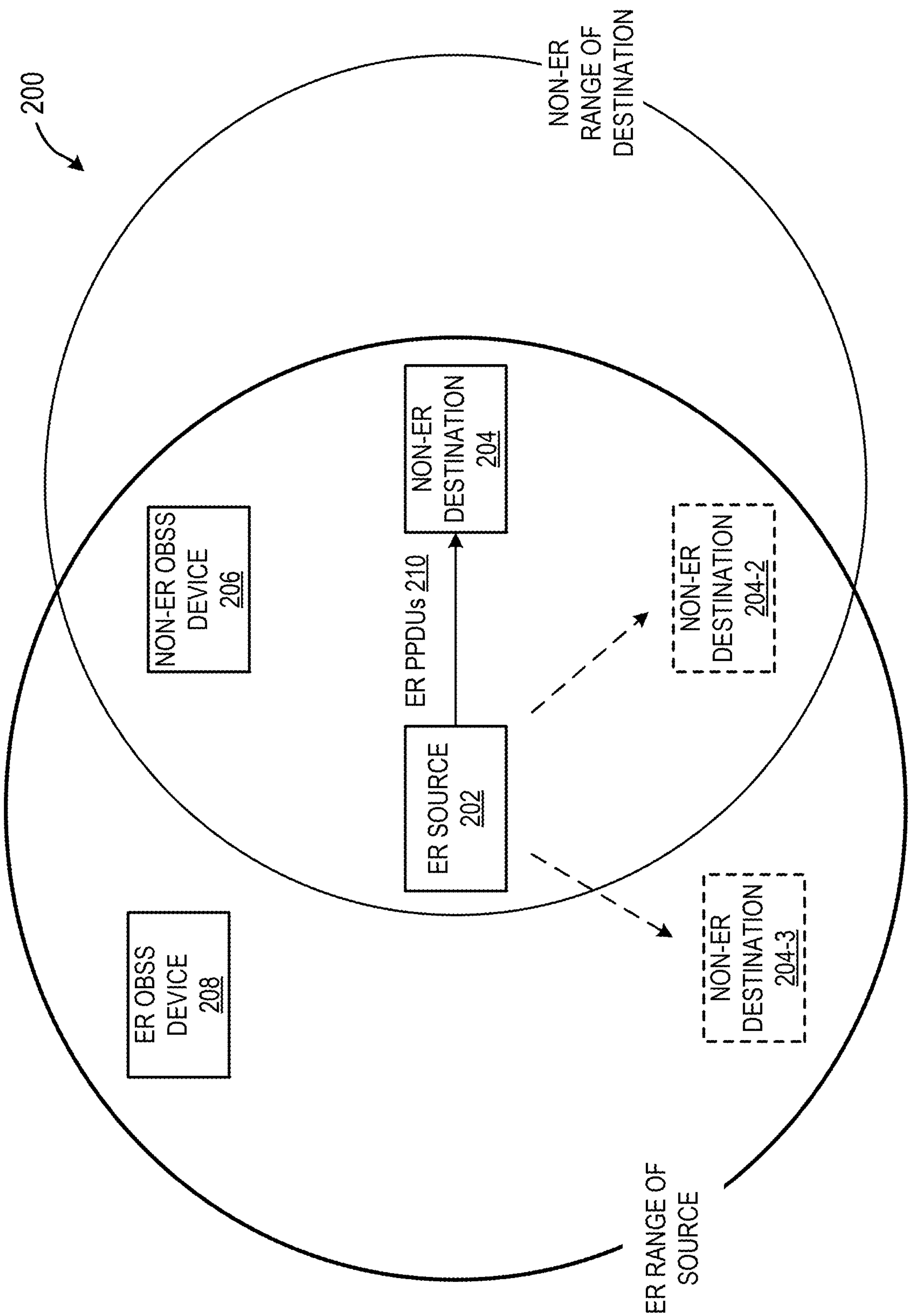

200
NON-ER RANGE OF DESTINATION
NON-ER OBSS DEVICE 206
NON-ER DESTINATION 204
ER PPDUs 210
ER SOURCE 202
NON-ER DESTINATION 204-2
NON-ER DESTINATION 204-3
ER OBSS DEVICE 208
ER RANGE OF SOURCE

FIG. 2A

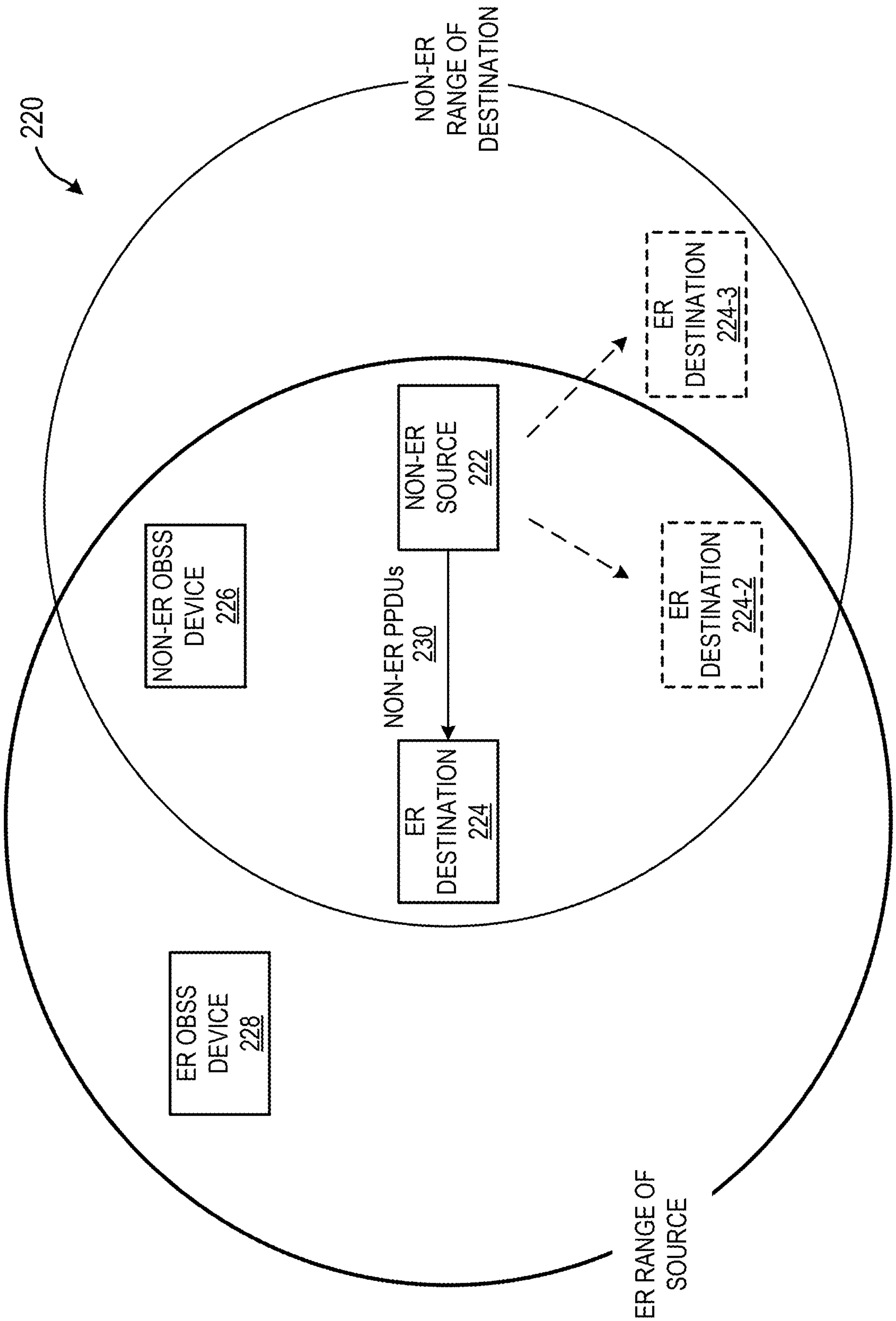
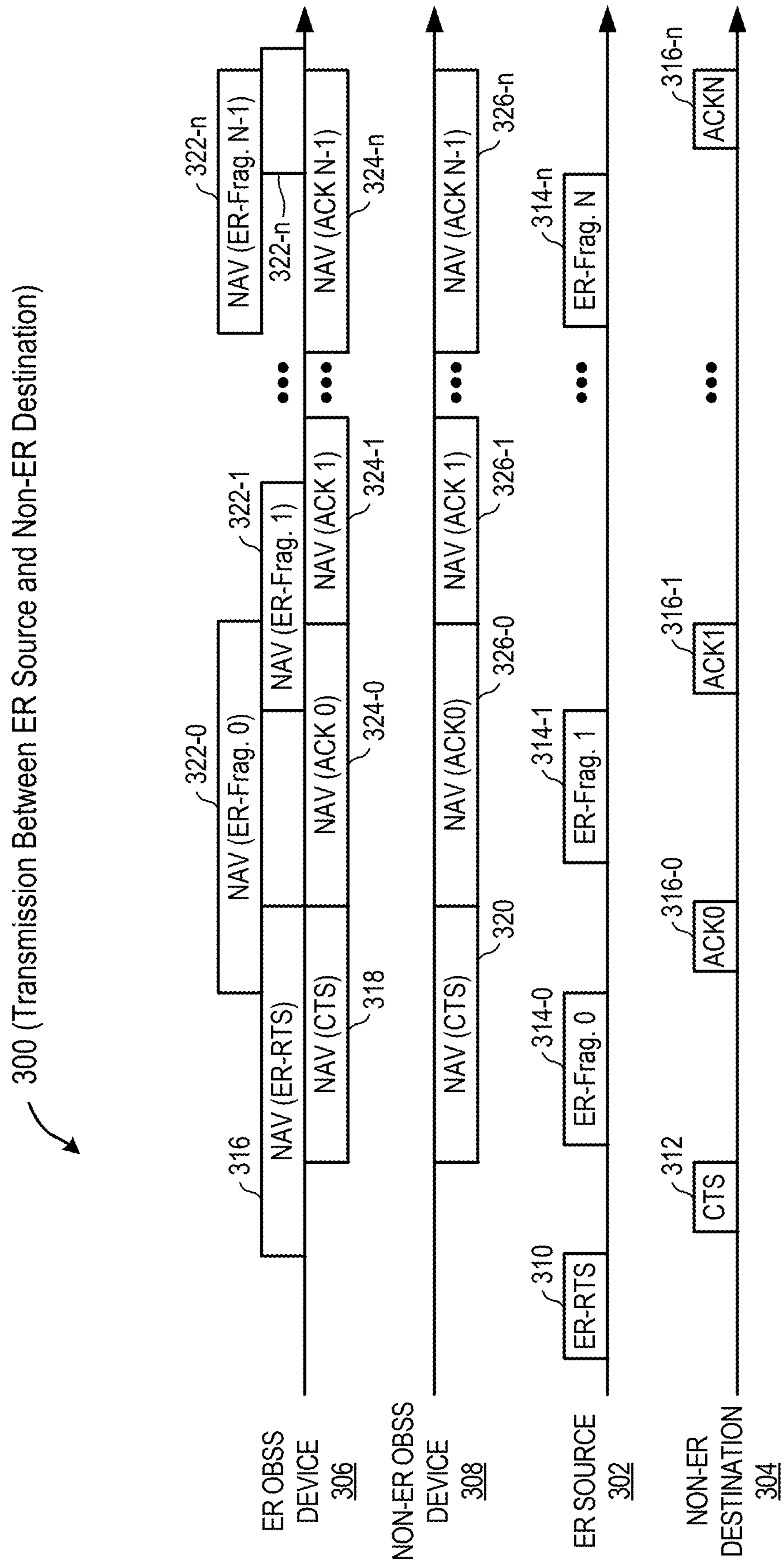

220
NON-ER RANGE OF DESTINATION
ER DESTINATION 224-3
NON-ER SOURCE 222
NON-ER OBSS DEVICE 226
NON-ER PPDUs 230
ER DESTINATION 224-2
ER DESTINATION 224
ER OBSS DEVICE 228
ER RANGE OF SOURCE

FIG. 2B



300 (Transmission Between ER Source and Non-ER Destination)
ER OBSS DEVICE 306
NON-ER OBSS DEVICE 308
ER SOURCE 302
NON-ER DESTINATION 304
NAV (ER-RTS)
316
NAV (CTS)
318
NAV (ER-Frag. 0)
322-0
NAV (ER-Frag. 1)
322-1
NAV (ACK 0)
324-0
NAV (ACK 1)
324-1
NAV (ER-Frag. N-1)
322-n
322-n
NAV (ACK N-1)
324-n
NAV (CTS)
320
NAV (ACK0)
326-0
NAV (ACK 1)
326-1
NAV (ACK N-1)
326-n
ER-RTS
310
ER-Frag. 0
314-0
ER-Frag. 1
314-1
ER-Frag. N
314-n
CTS
312
ACK0
316-0
ACK1
316-1
ACKN
316-n

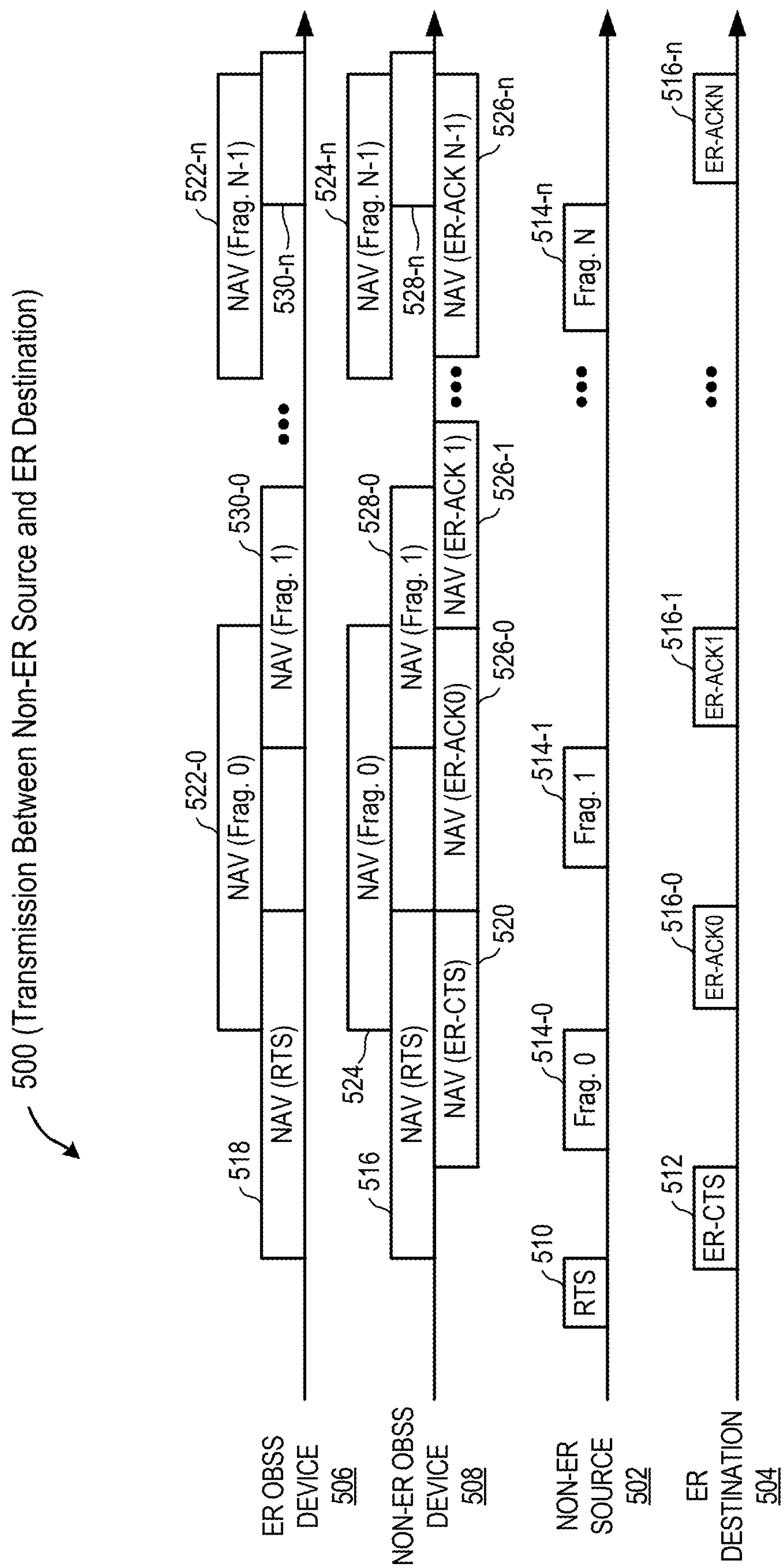
FIG. 3

400

STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	ER SOURCE sends ER-RTS	ER Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 0
2	Non-ER DESTINATION sends CTS	Non-ER and ER Rx supported devices in the vicinity of ER SOURCE. Upon receiving CTS, the ER SOURCE confirms the reservation and starts transmitting ER PPDU.	Until the end of ACK 0
3	ER SOURCE sends ER PPDU	ER Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 1
4	Non- ER DESTINATION sends ACK 0	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION	Until the end of ACK 1
--	CF-END	If ACK/CTS is not received at the ER SOURCE	Releases the channel for non-ER devices
--	ER-CF-END	If ACK/CTS is not received at the ER SOURCE. Sent SIFS duration after CF-END.	Releases the channel for ER devices

- Order of CF-end and ER-CF-end is interchangeable

FIG. 4



500 (Transmission Between Non-ER Source and ER Destination)
ER OBSS DEVICE 506
NON-ER OBSS DEVICE 508
NON-ER SOURCE 502
ER DESTINATION 504
518
NAV (RTS)
522-0
NAV (Frag. 0)
530-0
NAV (Frag. 1)
522-n
NAV (Frag. N-1)
530-n
516
524
NAV (RTS)
NAV (Frag. 0)
528-0
NAV (Frag. 1)
524-n
NAV (Frag. N-1)
528-n
NAV (ER-CTS)
520
NAV (ER-ACK0)
526-0
NAV (ER-ACK 1)
526-1
NAV (ER-ACK N-1)
526-n
510
RTS
514-0
Frag. 0
514-1
Frag. 1
514-n
Frag. N
512
ER-CTS
516-0
ER-ACK0
516-1
ER-ACK1
516-n
ER-ACKN

FIG. 5

600

STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	Non-ER SOURCE sends RTS	Non-ER and ER-Rx supported devices in vicinity of Non-ER SOURCE	Until the end of ER-ACK 0
2	ER DESTINATION sends ER-CTS	ER-Rx supported devices in the vicinity of DESTINATION. Upon receiving ER-CTS, the non-ER SOURCE confirms the reservation and starts transmitting non-ER-PPDU.	Until the end of ER-ACK 0
3	Non-ER SOURCE sends Non-ER PPDU	Non-ER and ER-Rx supported devices in vicinity of non-ER SOURCE	Until the end of ER-ACK 1
4	ER DESTINATION sends ER-ACK 0	ER-Rx supported devices in vicinity of ER DESTINATION	Until the end of ER-ACK 1
--	CF-END	If ER-ACK/ER-CTS is not received at the Non-ER SOURCE	Releases the channel for non-ER devices
--	ER-CF-END	If ER-ACK/ER-CTS is not received at the Non-ER SOURCE. Sent SIFS duration after CF-END.	Releases the channel for ER devices

- Order of CF-end and ER-CF-end is interchangeable

FIG. 6

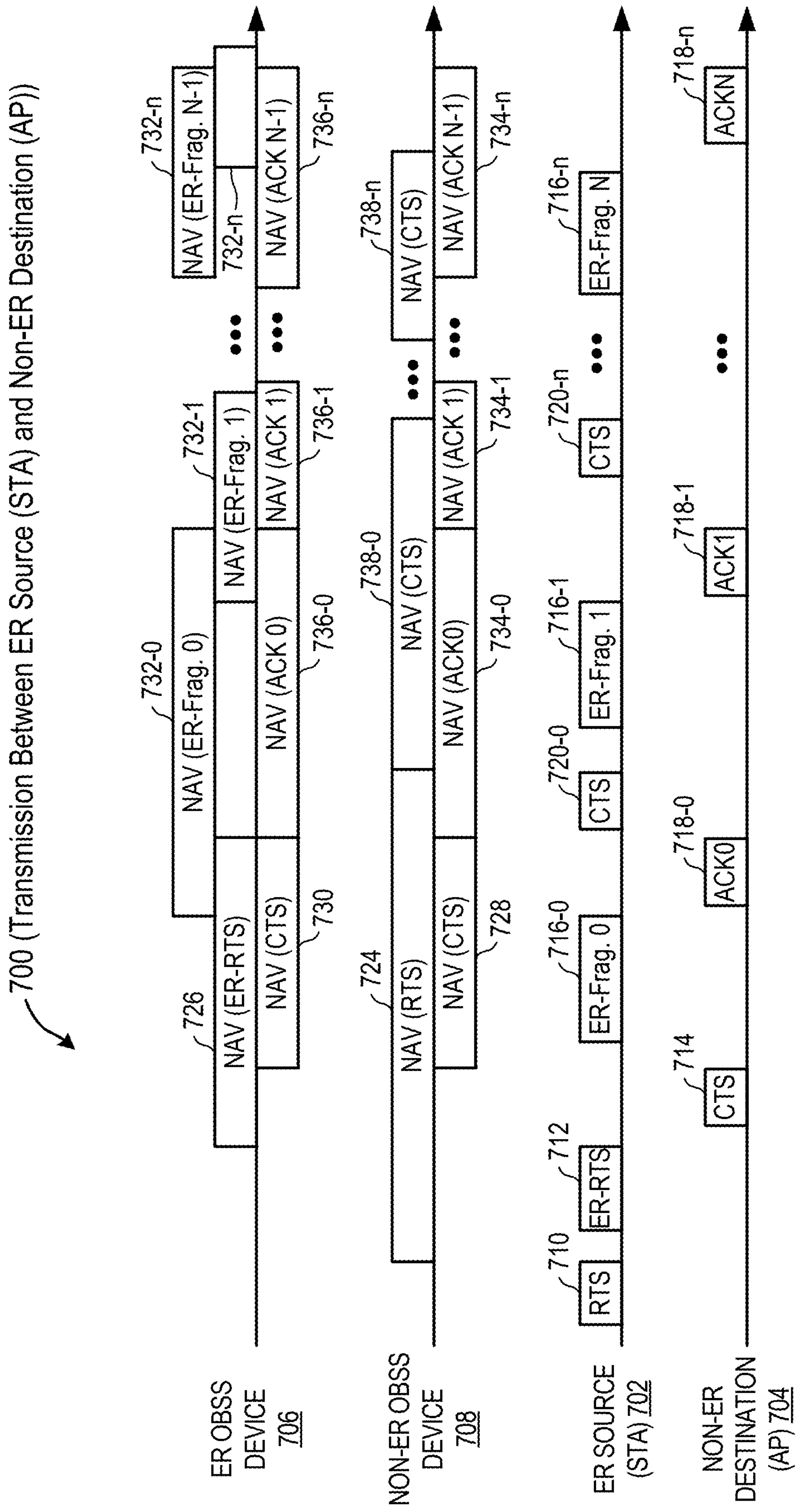

700 (Transmission Between ER Source (STA) and Non-ER Destination (AP))
ER OBSS DEVICE 706
NON-ER OBSS DEVICE 708
ER SOURCE (STA) 702
NON-ER DESTINATION (AP) 704
RTS
710
ER-RTS
712
CTS
714
ER-Frag. 0
716-0
ACK0
718-0
CTS
720-0
ER-Frag. 1
716-1
ACK1
718-1
CTS
720-n
ER-Frag. N
716-n
ACKN
718-n
NAV (RTS)
724
NAV (CTS)
728
NAV (CTS)
738-0
NAV (ACK0)
734-0
NAV (ACK 1)
734-1
NAV (CTS)
738-n
NAV (ACK N-1)
734-n
NAV (ER-RTS)
726
NAV (CTS)
730
NAV (ER-Frag. 0)
732-0
NAV (ACK 0)
736-0
NAV (ER-Frag. 1)
732-1
NAV (ACK 1)
736-1
NAV (ER-Frag. N-1)
732-n
NAV (ACK N-1)
736-n

FIG. 7

800

STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	ER SOURCE sends RTS-to-self	Non-ER supported devices in vicinity of ER SOURCE	Until the end of next CTS
2	ER SOURCE sends ER-RTS	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 0
3	Non-ER DESTINATION sends CTS	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION. Upon receiving CTS, the ER SOURCE confirms the reservation and starts transmitting ER-PPDU.	Until the end of ACK 0 (excludes the duration of ER-RTS of step 2 + SIFS + CTS of step 3)
4	ER SOURCE sends ER PPDU	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 1
5	Non-ER DESTINATION sends ACK 0	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION	Until the end of ACK 1
6	ER SOURCE sends CTS-to-self	Non-ER supported devices in vicinity of ER SOURCE	Until the end of next CTS
--	CF-END	If ACK/CTS is not received at the ER SOURCE	Releases the channel for non-ER devices
--	ER-CF-END	If ACK/CTS is not received at the ER SOURCE. Sent SIFS duration after CF-END.	Releases the channel for ER devices

- For single transaction, ER Source sends RTS-to-self to reserve the channel until ACK0
- Order of CF-end and ER-CF-end is interchangeable

FIG. 8

900

STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	ER SOURCE sends RTS-to-self	Non-ER supported devices in vicinity of ER SOURCE	Until the end of ACK 0
2	ER SOURCE sends ER-RTS	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 0
3	Non-ER DESTINATION sends CTS	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION. Upon receiving CTS, the ER SOURCE confirms the reservation and starts transmitting ER-PPDU.	Until the end of ACK 0 (excludes the duration of ER-RTS of step 2 + SIFS + CTS of step 3)
4	ER SOURCE sends ER PPDU	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 1
5	Non-ER DESTINATION sends ACK 0	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION	Until the end of ACK 1
--	CF-END	If ACK/CTS is not received at the ER SOURCE	Releases the channel for non-ER devices
--	ER-CF-END	If ACK/CTS is not received at the ER SOURCE. Sent SIFS duration after CF-END.	Releases the channel for ER devices

- Order of CF-end and ER-CF-end is interchangeable

FIG. 9

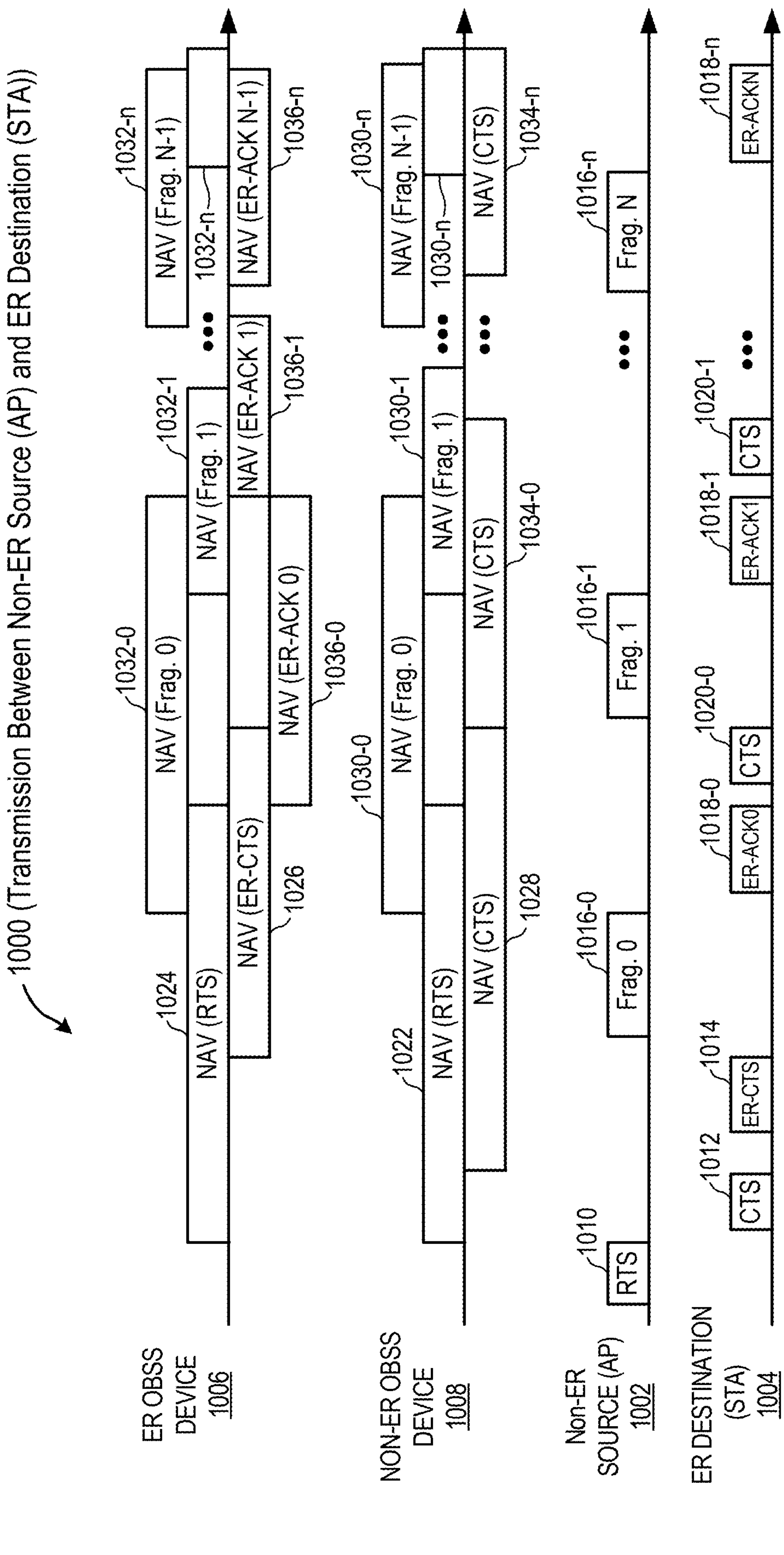

1000 (Transmission Between Non-ER Source (AP) and ER Destination (STA))
ER OBSS DEVICE 1006
NON-ER OBSS DEVICE 1008
Non-ER SOURCE (AP) 1002
ER DESTINATION (STA) 1004
NAV (RTS)
1024
NAV (ER-CTS)
1026
NAV (Frag. 0)
1032-0
NAV (ER-ACK 0)
1036-0
NAV (Frag. 1)
1032-1
NAV (ER-ACK 1)
1036-1
NAV (Frag. N-1)
1032-n
1032-n
NAV (ER-ACK N-1)
1036-n
NAV (RTS)
1022
NAV (CTS)
1028
NAV (Frag. 0)
1030-0
NAV (CTS)
1034-0
NAV (Frag. 1)
1030-1
NAV (Frag. N-1)
1030-n
1030-n
NAV (CTS)
1034-n
RTS
1010
Frag. 0
1016-0
Frag. 1
1016-1
Frag. N
1016-n
CTS
1012
ER-CTS
1014
ER-ACK0
1018-0
CTS
1020-0
ER-ACK1
1018-1
CTS
1020-1
ER-ACKN
1018-n

FIG. 10

1100

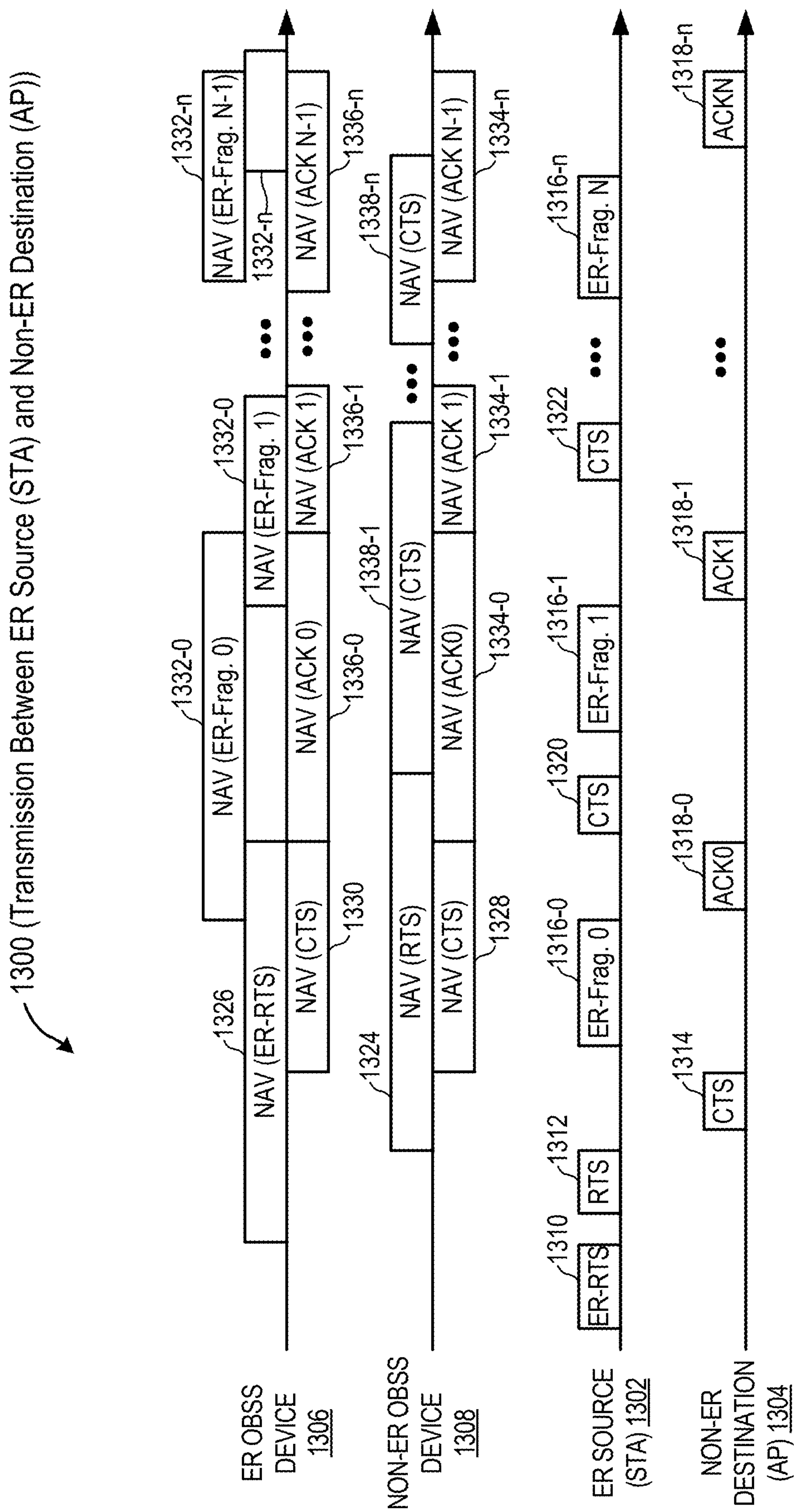
STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	Non-ER SOURCE sends RTS-to-self	Non-ER and ER-Rx supported devices in vicinity of Non-ER SOURCE	Until the end of next ER-ACK 0
2	ER DESTINATION sends CTS	Non-ER supported devices in vicinity of ER DESTINATION	Until the end of next CTS
3	ER DESTINATION sends ER-CTS	ER-Rx supported devices in vicinity of ER DESTINATION. Upon receiving ER-CTS, the non-ER SOURCE confirms the reservation and starts transmitting Non-ER-PPDU.	Until the end of ER-ACK 0 (excludes the duration of RTS of step 1 + 2*SIFS + CTS of step 2 + ER-CTS of step 3)
4	Non-ER SOURCE sends non-ER PPDU	Non-ER and ER-Rx supported devices in vicinity of Non-ER SOURCE	Until the end of ER-ACK 1
5	ER DESTINATION sends ER-ACK 0	ER-Rx supported devices in vicinity of ER DESTINATION	Until the end of ER-ACK 1
6	ER DESTINATION sends CTS-to-self	Non-ER supported devices in vicinity of ER DESTINATION	Until the end of next CTS
--	CF-END	If ER-ACK/ER-CTS is not received at the Non-ER SOURCE	Releases the channel for ER and non-ER devices

FIG. 11

1200

STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	Non-ER SOURCE sends RTS-to-self	Non-ER and ER-Rx supported devices in vicinity of Non-ER SOURCE	Until the end of next ER-ACK 0
2	ER DESTINATION sends CTS	Non-ER supported devices in vicinity of ER DESTINATION	Until the end of next ER-ACK 0
3	ER DESTINATION sends ER-CTS	ER-Rx supported devices in vicinity of ER DESTINATION. Upon receiving ER-CTS, the non-ER SOURCE confirms the reservation and starts transmitting Non-ER-PPDU.	Until the end of ER-ACK 0 (excludes the duration of RTS of step 1 + 2*SIFS + CTS of step 2 + ER-CTS of step 3)
4	Non-ER SOURCE sends non-ER PPDU	Non-ER and ER-Rx supported devices in vicinity of Non-ER SOURCE	Until the end of ER-ACK 1
5	ER DESTINATION sends ER-ACK 0	Non-ER and ER-Rx supported devices in vicinity of ER DESTINATION	Until the end of ER-ACK 1
--	CF-END	If ER-ACK/ER-CTS is not received at the Non-ER SOURCE	Releases the channel for ER and non-ER devices

FIG. 12



1300 (Transmission Between ER Source (STA) and Non-ER Destination (AP))
ER OBSS DEVICE 1306
NON-ER OBSS DEVICE 1308
ER SOURCE (STA) 1302
NON-ER DESTINATION (AP) 1304
NAV (ER-RTS)
1326
NAV (CTS)
1330
NAV (ER-Frag. 0)
1332-0
NAV (ACK 0)
1336-0
NAV (ER-Frag. 1)
1332-0
NAV (ACK 1)
1336-1
NAV (ER-Frag. N-1)
1332-n
1332-n
NAV (ACK N-1)
1336-n
NAV (RTS)
1324
NAV (CTS)
1328
NAV (CTS)
1338-1
NAV (ACK0)
1334-0
NAV (ACK 1)
1334-1
NAV (CTS)
1338-n
NAV (ACK N-1)
1334-n
ER-RTS
1310
RTS
1312
ER-Frag. 0
1316-0
CTS
1320
ER-Frag. 1
1316-1
CTS
1322
ER-Frag. N
1316-n
CTS
1314
ACK0
1318-0
ACK1
1318-1
ACKN
1318-n

FIG. 13

1400

STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	ER SOURCE sends ER-RTS	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 0
2	ER SOURCE sends RTS-to-self	Non-ER supported devices in vicinity of ER SOURCE	Until the end of next CTS (excludes the duration of ER-RTS of step 1 + SIFS + RTS of step 2)
3	Non-ER DESTINATION sends CTS	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION. Upon receiving CTS, the ER SOURCE confirms the reservation and starts transmitting ER-PPDU.	Until the end of ACK 0 (excludes the duration of ER-RTS of step 1 +RTS of step 2 + 2*SIFS + CTS of step 3)
4	ER SOURCE sends ER PPDU	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 1
5	Non-ER DESTINATION sends ACK 0	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION	Until the end of ACK 1
6	ER SOURCE sends CTS-to-self	Non-ER supported devices in vicinity of ER SOURCE	Until the end of next CTS
--	CF-END	If ACK/CTS is not received at the ER SOURCE	Releases the channel for non-ER devices
--	ER-CF-END	If ACK/CTS is not received at the ER SOURCE. Sent SIFS duration after CF-END.	Releases the channel for ER devices

- For single transaction, ER Source sends RTS-to-self to reserve the channel until ACK0
- Order of CF-end and ER-CF-end is interchangeable

FIG. 14

1500

STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	ER SOURCE sends ER-RTS	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 0
2	ER SOURCE sends RTS-to-self	Non-ER supported devices in vicinity of ER SOURCE	Until the end of ACK 0 (excludes the duration of ER-RTS of step 1 + SIFS + RTS of step 2)
3	Non-ER DESTINATION sends CTS	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION. Upon receiving CTS, the ER SOURCE confirms the reservation and starts transmitting ER-PPDU.	Until the end of ACK 0 (excludes the duration of ER-RTS of step 1 + RTS of step 2 + 2*SIFS + CTS of step 3)
4	ER SOURCE sends ER PPDU	Non-ER and ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 1
5	Non-ER DESTINATION sends ACK 0	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION	Until the end of ACK 1
--	CF-END	If ACK is not received at the ER SOURCE	Releases the channel for non-ER devices
--	ER-CF-END	If ACK is not received at the ER SOURCE. Sent SIFS duration after CF-END.	Releases the channel for ER devices

- Order of CF-end and ER-CF-end is interchangeable

FIG. 15

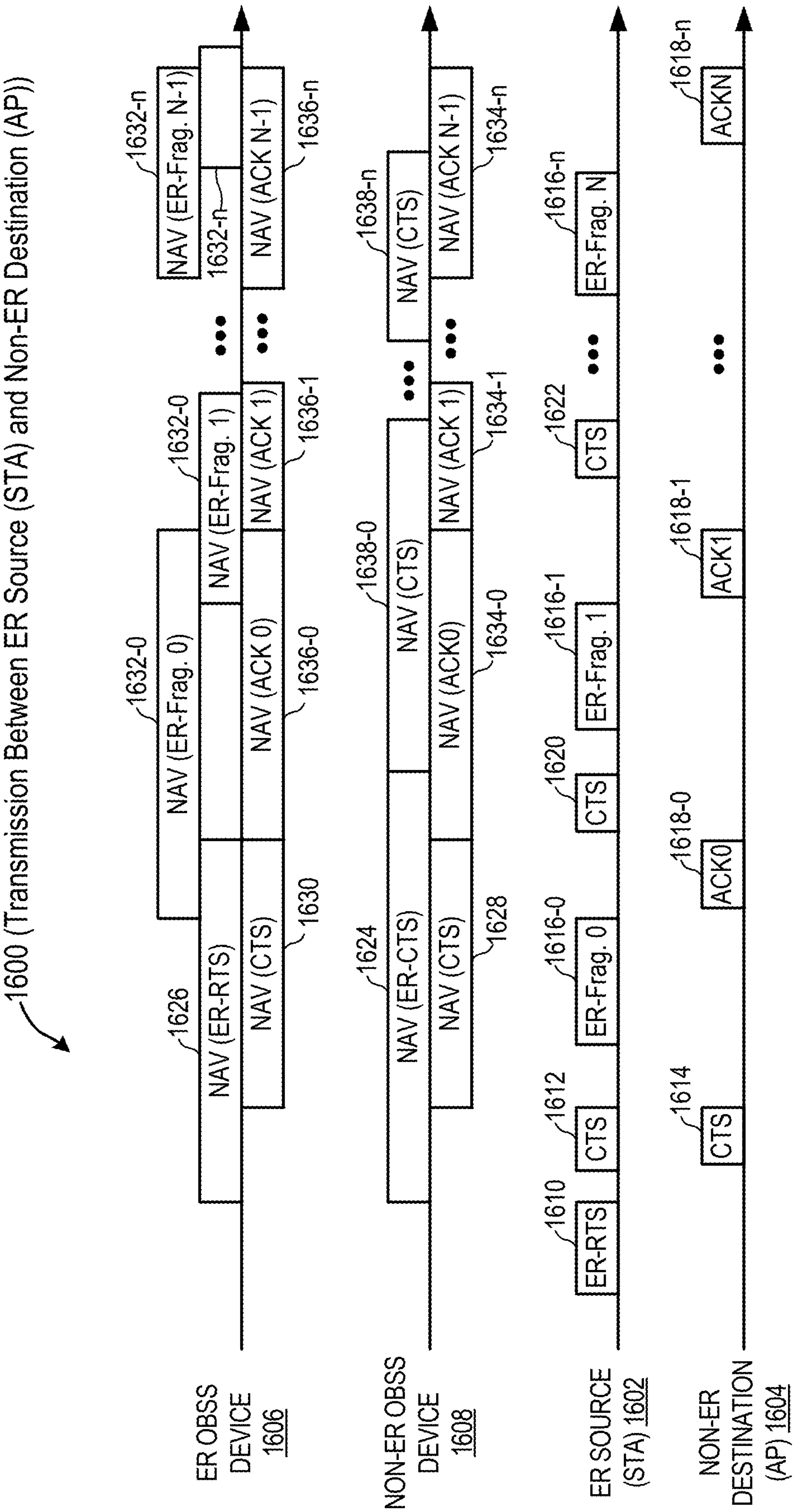

1600 (Transmission Between ER Source (STA) and Non-ER Destination (AP))
ER OBSS DEVICE 1606
NON-ER OBSS DEVICE 1608
ER SOURCE (STA) 1602
NON-ER DESTINATION (AP) 1604
NAV (ER-RTS)
1626
NAV (CTS)
1630
NAV (ER-Frag. 0)
1632-0
NAV (ER-Frag. 1)
NAV (ACK 0)
1636-0
NAV (ACK 1)
1636-1
NAV (ER-Frag. N-1)
1632-n
NAV (ACK N-1)
1636-n
NAV (ER-CTS)
1624
1628
NAV (CTS)
1638-0
NAV (ACK0)
1634-0
1634-1
1638-n
1634-n
ER-RTS
1610
CTS
1612
ER-Frag. 0
1616-0
1620
ER-Frag. 1
1616-1
1622
ER-Frag. N
1616-n
1614
ACK0
1618-0
ACK1
1618-1
ACKN
1618-n

FIG. 16

1700

STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	ER SOURCE sends ER-RTS	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 0
2	ER SOURCE sends CTS-to-self; Non-ER DESTINATION sends CTS with source address	Non-ER supported devices in vicinity of ER SOURCE; Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION.	Until the end of next CTS
3	ER SOURCE sends ER PPDU	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 1
4	Non-ER DESTINATION sends ACK 0	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION	Until the end of ACK 1
5	ER SOURCE sends CTF-to-self	Non-ER supported devices in vicinity of ER SOURCE	Until the end of next CTS
--	CF-END	If ACK is not received at the ER SOURCE	Releases the channel for non-ER devices
--	ER-CF-END	If ACK is not received at the ER SOURCE. Sent SIFS duration after CF-END.	Releases the channel for ER devices

- For single transaction, ER Source sends CTS-to-self and Non-ER Destination sends CTS to reserve the channel until ACK0
- For simultaneous CTS, TA address and RA address are same (either Source address or Destination address)
- Order of CF-end and ER-CF-end is interchangeable

FIG. 17

1800

STEP	PACKET SENT	CHANNEL RESERVATION ACROSS:	CHANNEL RESERVATION DURATION
1	ER SOURCE sends ER-RTS	ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 0
2	ER SOURCE sends CTS-to-self; Non-ER DESTINATION sends CTS with source address	Non-ER supported devices in vicinity of ER SOURCE; Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION.	Until the end of ACK 0 (excludes the duration of ER-RTS of step 1 + SIFS + CTS of step 2)
3	ER SOURCE sends ER PPDU	Non-ER and ER-Rx supported devices in vicinity of ER SOURCE	Until the end of ACK 1
4	Non-ER DESTINATION sends ACK 0	Non-ER and ER-Rx supported devices in vicinity of Non-ER DESTINATION	Until the end of ACK 1
--	CF-END	If ACK is not received at the ER SOURCE	Releases the channel for non-ER devices
--	ER-CF-END	If ACK is not received at the ER SOURCE. Sent SIFS duration after CF-END.	Releases the channel for ER devices

- For simultaneous CTS, TA address and RA address are same (either Source address or Destination address)
- Order of CF-end and ER-CF-end is interchangeable

FIG. 18

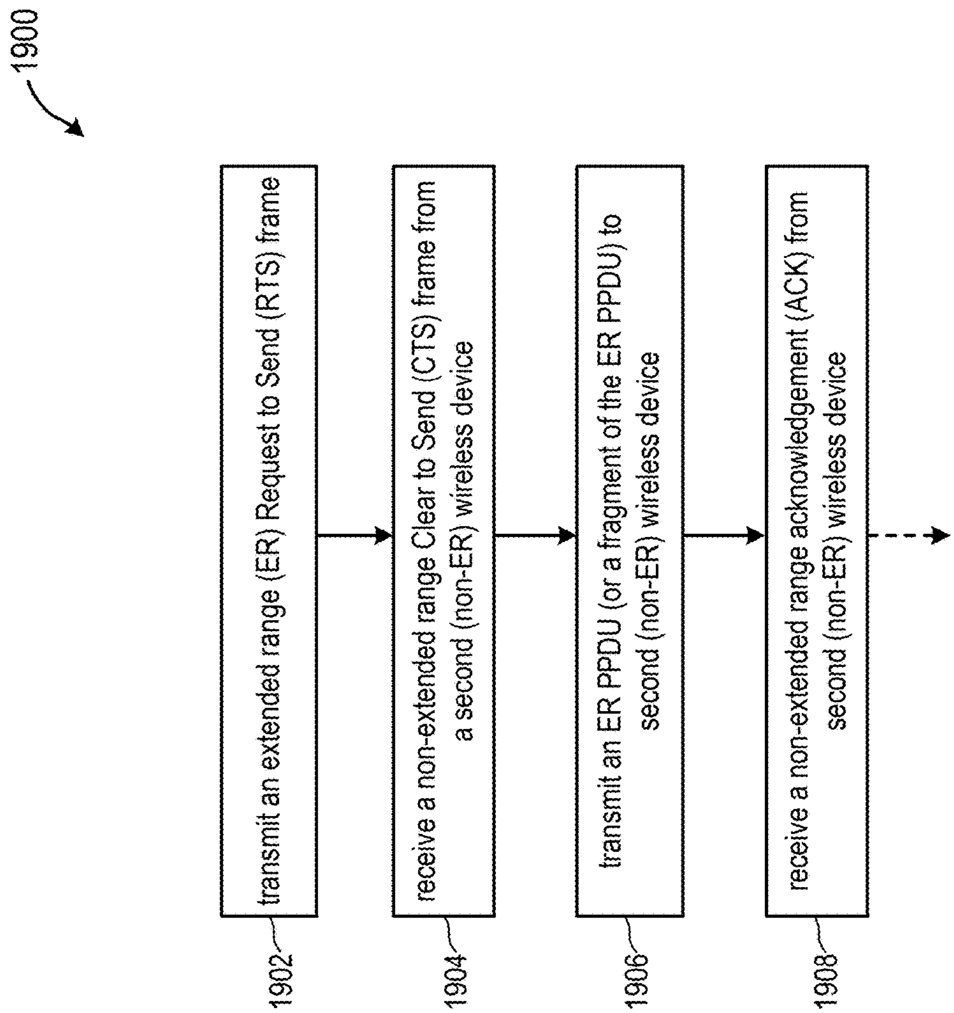

1900
transmit an extended range (ER) Request to Send (RTS) frame
1902
receive a non-extended range Clear to Send (CTS) frame from a second (non-ER) wireless device
1904
transmit an ER PPDU (or a fragment of the ER PPDU) to second (non-ER) wireless device
1906
receive a non-extended range acknowledgement (ACK) from second (non-ER) wireless device
1908

FIG. 19

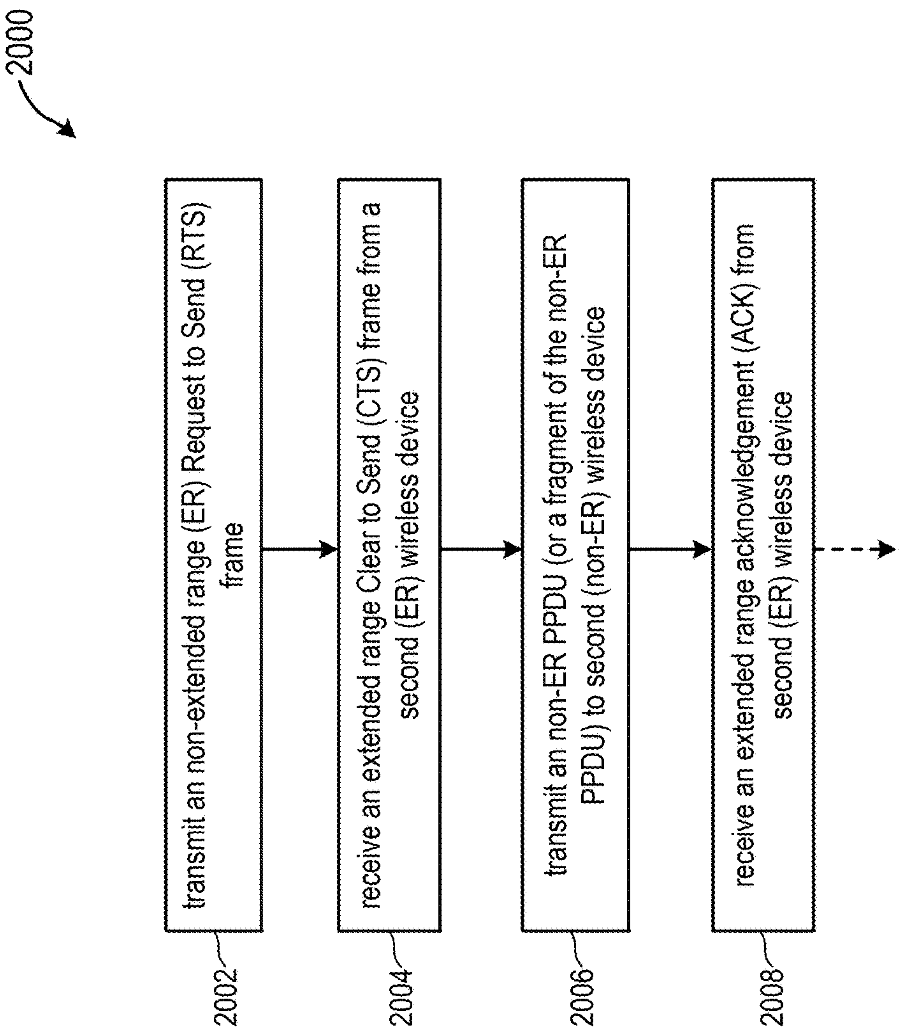

2000
transmit an non-extended range (ER) Request to Send (RTS) frame
2002
receive an extended range Clear to Send (CTS) frame from a second (ER) wireless device
2004
transmit an non-ER PPDU (or a fragment of the non-ER PPDU) to second (non-ER) wireless device
2006
receive an extended range acknowledgement (ACK) from second (ER) wireless device
2008

FIG. 20

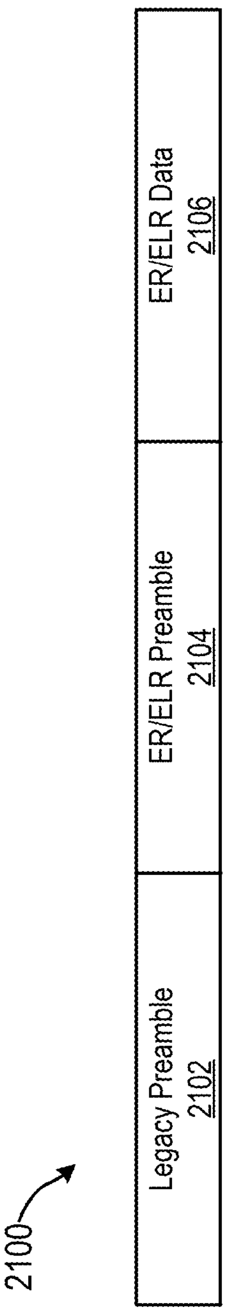

2100
Legacy Preamble
2102
ER/ELR Preamble
2104
ER/ELR Data
2106

FIG. 21

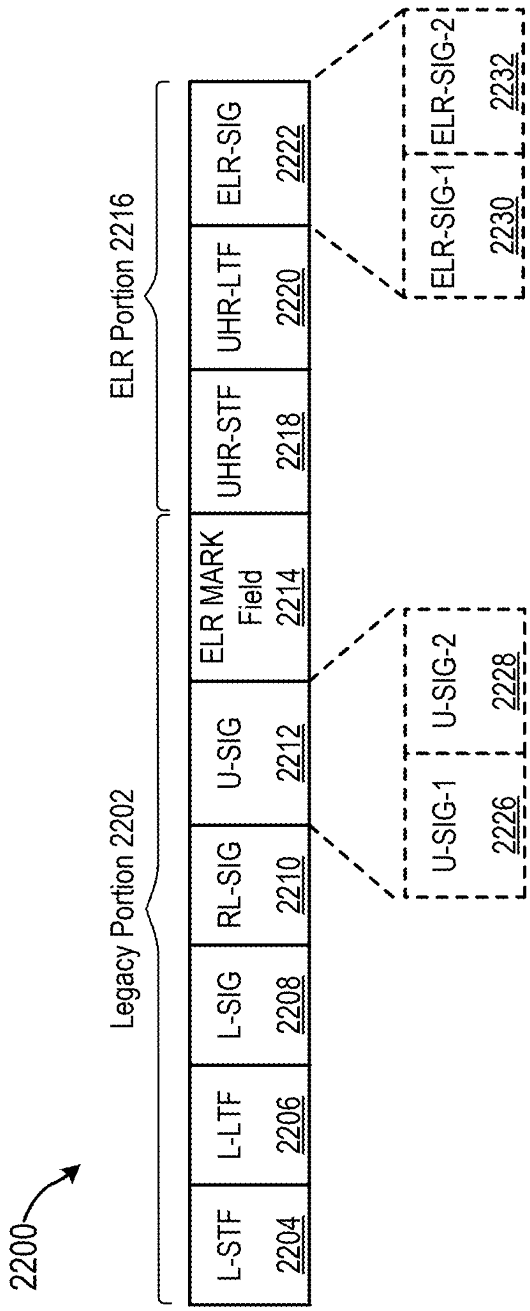

2200
Legacy Portion 2202
ELR Portion 2216
L-STF
2204
L-LTF
2206
L-SIG
2208
RL-SIG
2210
U-SIG
2212
ELR MARK
Field
2214
UHR-STF
2218
UHR-LTF
2220
ELR-SIG
2222
U-SIG-1
2226
U-SIG-2
2228
ELR-SIG-1
2230
ELR-SIG-2
2232

FIG. 22

RTS-CTS PROCEDURE FOR EXTENDED RANGE PACKET TRANSMISSION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present U.S. Utility Patent Application claims priority pursuant to 35 U.S.C. § 119(a) to Indian Provisional Patent Application Serial Number 202441014419, entitled "RTS-CTS PROCEDURE FOR EXTENDED RANGE PACKET TRANSMISSION WITH AP HAVING MORE RANGE THAN STA", filed Feb. 28, 2024, and Indian Provisional Patent Application Serial Number 202441014478, entitled "RTS-CTS PROCEDURE FOR EXTENDED RANGE PACKET UPLINK TRANSMISSION WITH AP HAVING MORE RANGE THAN STA", filed Feb. 28, 2024, the contents of both of which are incorporated herein by reference in their entirety and made part of the present U.S. Utility Patent Application for all purposes.

BACKGROUND

technical field

[0002] This disclosure relates generally to wireless communications, and more specifically to reservation of a transmission channel.

Description of Related Art

[0003] Wireless local area networks (WLANs) have evolved rapidly over the past couple of decades, including WLANs that conform to the Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards. A typical 802.11-based WLAN is formed by one or more access points (APs) that provide a shared wireless communication medium for servicing a number of client devices or stations (STAs). In particular, an AP manages a Basic Service Set (BSS) that is identified by a Basic Service Set Identifier (BSSID) and advertised by the AP. The AP periodically broadcasts beacon frames to enable STAs within wireless range of the AP to establish and maintain communication links with the AP.

[0004] In such WLANs, wireless devices such as Access Points (APs) and client stations (STAs) wirelessly transmit and receive physical layer protocol data units (PPDUs). As various new services and deployment scenarios are supported by these wireless devices, the devices may need to be able to transmit and receive signals over longer ranges. To extend the range that the PPDUs are transmitted and received, the IEEE 802.11ax and IEEE 802.11be amendments to the IEEE 802.11 standard define an extended range PPDU. The IEEE 802.11b amendment also describes direct sequence spread spectrum (DSSS) communications to support an extended range, but practical applications are limited due to low data rates.

[0005] The IEEE 802.11 standard further includes a Request to Send/Clear to Send (RTS/CTS) mechanism intended to reduce frame collisions and manage wireless medium access. In conventional operation, when a STA/AP wants to transmit data, it first sends an RTS frame to the intended recipient. The RTS frame includes information about the duration of the proposed data transmission and any subsequent acknowledgements (ACKs). Upon receiving the RTS frame, the recipient waits for a Short Interframe Space (SIFS) period and then responds with a CTS frame. The CTS frame repeats the duration information, thereby reserving the wireless medium for the specified time. Once the sender receives the CTS frame, it proceeds to send the actual data frames. Other stations in a BSS that overhear the RTS or CTS frames may set their Network Allocation Vector (NAV) timers to defer their transmissions for the duration specified in the RTS/CTS frames.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] One or more embodiments will now be described by way of example only with reference to the accompanying drawings, in which:

[0007] FIG. **1** illustrates an example of a wireless local area network (WLAN) in accordance with embodiments of the present disclosure;

[0008] FIG. **2**A illustrates an example of a WLAN including an extended range source device communicating with a non-extended range destination device in accordance with embodiments of the present disclosure;

[0009] FIG. **2**B illustrates an example of a WLAN including a non-extended range source device communicating with an extended range destination device in accordance with embodiments of the present disclosure;

[0010] FIG. **3** illustrates a first channel reservation scheme in accordance with an embodiment of the present disclosure;

[0011] FIG. **4** illustrates a table presenting a stepwise summarization of the first channel reservation scheme in accordance with an embodiment of the present disclosure;

[0012] FIG. **5** illustrates a second channel reservation scheme in accordance with an embodiment of the present disclosure;

[0013] FIG. **6** illustrates a table presenting a stepwise summarization of the second channel reservation scheme in accordance with an embodiment of the present disclosure;

[0014] FIG. **7** illustrates a third channel reservation scheme in accordance with an embodiment of the present disclosure;

[0015] FIG. **8** illustrates a table presenting a stepwise summarization of the third channel reservation scheme in accordance with an embodiment of the present disclosure;

[0016] FIG. **9** illustrates a table presenting a stepwise summarization of a variation of the third channel reservation scheme in accordance with an embodiment of the present disclosure;

[0017] FIG. **10** illustrates a fourth channel reservation scheme in accordance with an embodiment of the present disclosure;

[0018] FIG. **11** illustrates a table presenting a stepwise summarization of the fourth channel reservation scheme in accordance with an embodiment of the present disclosure;

[0019] FIG. **12** illustrates a table presenting a stepwise summarization of a variation of the fourth channel reservation scheme in accordance with an embodiment of the present disclosure;

[0020] FIG. **13** illustrates a fifth channel reservation scheme in accordance with an embodiment of the present disclosure;

[0021] FIG. **14** illustrates a table presenting a stepwise summarization of the fifth channel reservation scheme in accordance with an embodiment of the present disclosure;

[0022] FIG. **15** illustrates a table presenting a stepwise summarization of a variation of the fifth channel reservation scheme in accordance with an embodiment of the present disclosure;

[0023] FIG. **16** illustrates a sixth channel reservation scheme in accordance with an embodiment of the present disclosure;

[0024] FIG. **17** illustrates a table presenting a stepwise summarization of the sixth channel reservation scheme in accordance with an embodiment of the present disclosure;

[0025] FIG. **18** illustrates a table presenting a stepwise summarization of a variation of the sixth channel reservation scheme in accordance with an embodiment of the present disclosure;

[0026] FIG. **19** is a flow diagram illustrating an example method for channel reservation in accordance with an embodiment of the present disclosure;

[0027] FIG. **20** is a flow diagram illustrating another example method for channel reservation in accordance with an embodiment of the present disclosure;

[0028] FIG. **21** illustrates an example of an Extended Range (ER)/Enhanced Long Range (ELR) physical layer protocol data unit (PPDU); and

[0029] FIG. **22** illustrates an example of non-legacy ELR PPDU.

DETAILED DESCRIPTION

[0030] The various implementations described in the following description relate generally to new and innovative channel reservation schemes applicable to scenarios where a wireless source device(s) and a wireless destination device (s) have differing transmission ranges. More particularly, channel reservation schemes are described for transmissions (e.g., uplink transmissions) between (1) an extended range (ER) source and a non-ER destination and (2) a non-ER source and an ER destination. In various examples, request to send/clear to send (RTS/CTS) mechanisms are described using both ER frames and non-ER frames to reserve a channel for extended range in scenarios wherein there may be asymmetry between the transmission or reception ranges of an AP and a STA. For example, the described channel reservations schemes may include an ER transmitter/source transmitting certain RTS and/or CTS frames in an extended range (ER)/Enhanced Long Range (ELR) physical layer protocol data unit (PPDU) format.

[0031] As used herein, the term "non-legacy" may refer to PPDU formats and communication protocols conforming with the IEEE 802.11bn amendment to the IEEE 802.11 standard (also referred to as "802.11bn", "UHR" or "Wi-Fi 8") as well as future generations/amendments. In contrast, the term "legacy" may be used herein to refer to PPDU formats and communication protocols conforming to the IEEE 802.11be (also referred to as Extremely High Throughput or "EHT" or "Wi-Fi 7") or IEEE 802.11ax (also referred to as High Efficiency or "HE" or "Wi-Fi 6/6E") amendments to the IEEE 802.11 standard, or earlier generations of the IEEE 802.11 standard, but not conforming to all mandatory features of 802.11bn or future generations of the IEEE 802.11 standard. In some implementations, the channel reservation schemes described herein may support multiple versions of the IEEE 802.11 standard.

[0032] As used herein, the term ER PPDU refers to an "Extended Range" PPDU (e.g., such as defined by 802.11ax) used to transmit data over longer distances within a standard WLAN, and the term ELR PPDU refers to newer "Extended Long Range" formats designed to achieve even greater range (potentially at the cost of data throughput). The terms "ER PPDU" and "ELR PPDU" are used interchangeably herein unless specifically noted otherwise.

[0033] Further, an AP and (associated or unassociated) STA of a described embodiment may each support extended range signaling, but the effective range of one may be greater than the other due to differing implementations of an amendment to the IEEE 802.11 standard (e.g., support for optional features, componentry, etc.) or compliance with differing amendments to the IEEE 802.11 standard (legacy or non-legacy) that define different ER capabilities affording differing extended range capabilities. In such instances, the terms "non-ER AP" and "non-ER STA" as used herein can refer to the AP or STA having a relatively lesser effective range. In addition, the terms "channel", "transmission channel", "wireless channel" and "wireless medium" are synonymous and may be used interchangeably herein.

[0034] Particular implementations of the subject matter described in the present disclosure can be implemented to realize one or more of the following potential advantages. By providing effective procedures for minimizing collisions and interference in wireless networks involving devices having disparate transmission ranges, the described channel reservation schemes support gains in overall network throughput that will be achievable in accordance with the IEEE 802.11bn amendment of the IEEE 802.11 standard, improvements to the management of time-sensitive traffic (e.g., traffic with bounded low latency), and mitigation of potential interference levels.

[0035] FIG. **1** illustrates an example of a wireless local area network (WLAN) **100** in accordance with embodiments of the present disclosure. The illustrated WLAN includes a wireless access point (AP) **102** and one or more wireless client stations (STAs) **116** (e.g., **116-1**, **116-2**, and **116-3**). The AP**102** of this example is configured to transmit downlink Extended Range (ER) and/or non-ER PPDUs and receive uplink ELR PPDUs and/or non-ER PPDUs. The ER PPDUs can have a format and contents such as described in greater detail below with reference to FIG. **21** and FIG. **22**.

[0036] The illustrated AP **102** includes a host processor **104** coupled to a network interface **106**. The network interface **106** includes a medium access control (MAC) processing unit **108** and a physical layer (PHY) processing unit **110**. The PHY processing unit **110** includes a plurality of transceivers **112-1**, **112-2** and **112-3** (e.g., transmitters and/or receivers) coupled to a respective plurality of antennas **114-1**, **114-2** and **114-3**. Although three transceivers **112** and three antennas **114** are illustrated in FIG. **1**, in other embodiments the AP **102** includes other suitable numbers (e.g., 1, 2, 4, 5, etc.) of transceivers **112** and antennas **114** in other embodiments. In one embodiment, the MAC processing unit **108** and the PHY processing unit **110** are configured to operate in compliance with the IEEE 802.11bn amendment to the IEEE 802.11 standard.

[0037] The illustrated WLAN **100** also includes one or more wireless client stations **116**. Three client stations **116** shown as **116-1**, **116-2**, and **116-3** are illustrated in FIG. **1**, but the WLAN **100** may include other suitable numbers (e.g., 1, 2, 3, 5, 6, etc.) of client stations **116** in various scenarios and embodiments. At least one of the client stations **116** (e.g., client station **116-1**) may be configured to operate in compliance with the IEEE 802.11bn amendment to the IEEE 802.11 standard to communicate with the AP **102**.

[0038] The client station **116-1** includes a host processor **118** coupled to a network interface **120** which includes a MAC processing unit **122** and a PHY processing unit **124**. The PHY processing unit **124** includes a plurality of transceivers **126-1**, **126-2** and **126-3**, and the transceivers **126** are coupled to a respective plurality of antennas **128-1**, **128-2** and **128-3**. Although three transceivers **126** and three antennas **128** are illustrated in FIG. **1**, the client station **116-1** includes other suitable numbers (e.g., 1, 2, 4, 5, etc.) of transceivers **126** and antennas **128** in other embodiments.

[0039] In various embodiments, the PHY processing unit **110** of the AP **102** is configured to generate and transmit (downlink) data units via the antenna(s) **144** over an air interface and the PHY processing unit **124** of the client station **116-1** is configured to receive the (downlink) data units via the antenna(s) **128** over the air interface. Similarly, the PHY processing unit **110** of the client station **116-1** is configured to generate and transmit (extended range and/or non-extended range) data units via the antenna(s) **128** and the PHY processing unit **110** of the AP **102** is configured to receive the (uplink) data units via the antenna(s) **124**. In an example, the data units may be physical layer data units (PPDUs) for communicating data between the AP **102** and the client station **116-1** and the PPDUs (and fields therein) may be transmitted as a waveform in a downlink or uplink direction.

[0040] In embodiments, the network interface **106** of the AP **102** and the network interface **120** of one or more of the client stations **116** are configured to generate, transmit and receive ER/ELR PPDUs having an extended range format to increase a range and/or a signal-to-noise (SNR) ratio associated with transmitting, receiving, and successfully decoding the ER/ELR PPDUs exchanged in the WLAN **100**. In an example, the ELR PPDUs are compliant with the IEEE 802.11bn (or later) amendment to the IEEE 802.11 standard, and include a legacy portion with legacy fields of one or more legacy IEEE 802.11 standards for backwards compatibility with legacy devices and an extended range (ER) portion with non-legacy fields of a non-legacy IEEE 802.11 standard which can be decoded by non-legacy devices. In an example, various of the fields of an ER PPDU are repeated in a time domain and/or duplicated in a frequency domain to increase a range and/or SNR associated with transmission and reception of data in the ER portion of the ER PPDUs. In another example, the ER PPDU may be a trigger frame which is transmitted by the AP device **102** and which is received by the client station **116** in a downlink direction.

[0041] The range extension features of the ER PPDU may allow a client station **116** to decode the ER portion of the ER PPDU at an extended range. Decoding is a process of determining a valid pattern of bits of the received ER PPDU referred to as decoded bits. In an example, the decoding may involve performing a parity check or CRC which determines whether the decoding is successful or is not successful. A downlink ER PPDU transmitted by AP **102** may solicit a response from a client station **116** in the form of an uplink ER PPDU or a non-ER PPDU.

[0042] In an embodiment, when operating in single-user mode, the AP **102** transmits a data unit to a single client station (DL SU transmission), or receives a data unit transmitted by a single client station (UL SU transmission), without simultaneous transmission to, or by, any other client station. When operating in multi-user mode, the AP **102** transmits a data unit that includes multiple data streams for multiple client stations (DL MU transmission), or receives data units simultaneously transmitted by multiple client stations (UL MU transmission). For example, in multi-user mode, a data unit transmitted by the MLD includes multiple data streams simultaneously transmitted by the AP **102** to respective client stations using respective spatial streams allocated for simultaneous transmission to the respective client stations and/or using respective sets of OFDM tones corresponding to respective frequency sub-channels allocated for simultaneous transmission to the respective client stations. In a further example, the AP **102** and/or client station(s) **116** may be configured as a multi-link device (MLD). In another example, the AP **102** and/or one or more of the client stations **116** are configured to transmit and receive PPDUs over a plurality of wireless links, including one or more of a 2.4 Gigahertz (GHz) link, a 5 GHz link, a 6 GHz link, and a mmWave link (e.g., a 45 GHz link and/or a 60 GHz link).

[0043] In an example, the illustrated AP **102** may be connected to a distribution system (DS) through a distribution system medium (DSM). The distribution system may be a wired network or a wireless network that is connected to a backbone network such as the Internet. The DSM may be a wired medium (e.g., Ethernet cables, telephone network cables, or fiber optic cables) or a wireless medium (e.g., infrared, broadcast radio, cellular radio, or microwaves). Although some examples of the DSM are described, the DSM is not limited to the examples described herein. In another example, the AP **102** and/or client stations **116** may be implemented in a laptop, a desktop personal computer (PC), a mobile phone, or other communications device that supports at least one WLAN communications standard (e.g., at least one IEEE 802.11 standard).

[0044] In an example, one or more of the AP **102** and client stations **116** may be implemented with circuitry such as one or more of analog circuitry, mixed signal circuitry, memory circuitry, logic circuitry, and processing circuitry that executes code stored in a memory that when executed by the processing circuitry performs the disclosed functions. For example, the AP **102** and client stations **116** may include memory storing operational instructions (software, program instructions, computer instructions, etc.) and one or more processing modules/processors, operably coupled to one or more wireless transceivers and the memory, configured to execute the operational to perform the described channel reservation schemes.

[0045] In another example, a network interface **106/120** includes one or more integrated circuit (IC) devices. In this example, at least some of the functionality of a MAC processing unit **108/122** and at least some of the functionality of the PHY processing unit **110** can be implemented on a single IC device. As another example, at least some of the functionality of the MAC processing unit **108** is implemented on a first IC device, and at least some of the functionality of the PHY processing unit **110** is implemented on a second IC device.

[0046] FIG. **2**A illustrates an example of a WLAN **200** including an extended range (ER) source **202** (e.g., a STA) communicating with a non-extended range (non-ER) destination(s) **204** (e.g., an AP) in accordance with embodiments of the present disclosure. In the illustrated example, the

WLAN **200** further includes a non-ER OBSS (Overlapping BSS) device **206** and an ER OBSS device **208** which may utilize one or more of the same transmission channels as the ER source **202** and non-ER destination **204**. One or more of the ER source **202**, non-ER destination(s) **204**, non-ER OBSS device **206** and ER OBSS device **208** may be an example of an AP **102** or STA **116** of FIG. **1**.

[0047] In the illustrated example, the ER source **202** may transmit ER PPDUs **210** to the non-ER destination **204**. In this example, one or more non-ER OBSS devices **206** are located within the non-ER range of the non-ER destination (s) **204**, one or more ER OBSS devices **208** and the ER destination(s) **204** are located within the ER range of the ER source **202**, and the ER source **202** is located within the non-ER range of the ER destination **204**. In the topology of the depicted WLAN **200**, the ER source **202**, the ER destination **204**, the non-ER OBSS device **206**, and the ER OBSS device **208** may correspond to the similarly labeled devices illustrated in channel reservation schemes of FIG. **3**, FIG. **7**, FIG. **13**, and FIG. **16**.

[0048] FIG. **2**B illustrates an example of a WLAN **220** including a non-ER source **222** (e.g., an AP) communicating with an ER destination(s) **224** (e.g., a STA(s)) in accordance with embodiments of the present disclosure. In the illustrated example, the WLAN **220** further includes a non-ER OBSS device **226** and an ER OBSS device **228** which may utilize one or more of the same transmission channels as the non-ER source **222** and ER destination **224**. One or more of the non-ER source **222**, ER destination(s) **224**, non-ER OBSS device **226** and ER OBSS device **228** may be an example of an AP **102** or STA **116** of FIG. **1**.

[0049] In the illustrated example, the non-ER source **222** may transmit non-ER PPDUs **230** to the ER destination **224**. In this example, the non-ER OBSS device **226** and the ER OBSS device **228** are located within the non-ER range of the non-ER source **222**, the non-ER OBSS device **226** and the ER OBSS device **228** are located within the ER range of the ER destination **224**, and the ER destination **224** is located within the non-ER range of the ER source **222**. In the topology of the depicted WLAN **220**, the non-ER source **222**, the ER destination **224**, the non-ER OBSS device **226**, and the ER OBSS device **228** may correspond to the similarly labeled devices illustrated in channel reservation schemes of FIG. **5** and FIG. **10**.

[0050] FIGS. **3-18** illustrate various channel reservation schemes in which RTS/CTS frames are exchanged using a combination of using both ER packets/frames and non-ER packets/frames to reserve a channel for extended range in scenarios wherein there may be asymmetry between the transmission or reception ranges of a source device and destination device. In various of the Figures, operation of Network Allocation Vector (NAV) timers is illustrated. Briefly, a NAV is a virtual carrier-sensing mechanism used in wireless networking protocols such as IEEE 802.11 to help manage access to a wireless medium/transmission channel. In operation, a NAV functions as a timer to indicate the duration for which a transmission channel will be occupied. In an example, when a STA receives a frame addressed to another device, it decodes a duration field in the frame header which specifics a time (e.g., in microseconds) transmission and any subsequent acknowledgements. The STA then sets a NAV timer to this value, during which it refrains from attempting to access the transmission channel. A STA/AP may maintain multiple NAV timers, including a NAV timer(s) that handles channel reservations for frames received from an Overlapping BSS (OBSS) that utilizes the same transmission channel.

[0051] FIG. **3** illustrates a first channel reservation scheme **300** in accordance with an embodiment of the present disclosure. In this example, an extended range (ER) source **302** may be required to communicate with a non-ER destination **304**. To facilitate such communication between the ER source **302** and the non-ER destination **304**, a transmission channel between the two devices may be reserved. When the transmission channel is reserved for the ER source **302** and the non-ER destination **304**, other devices within range may perceive the transmission channel as unavailable for use.

[0052] In the first channel reservation scheme **300**, the ER source transmits an extended range Request-to-Send (ER-RTS) frame **310** to non-ER destination **304**. The ER-RTS frame **310** contains a source address, a destination address, and a transaction duration. The transaction duration may indicate, for example, a time required for completion of the all of the related transactions between the ER source **302** and the non-ER destination **304**. In the illustrated example, an ER OBSS device **306** detects the ER-RTS frame **310** and sets or updates a corresponding NAV timer (or “NAV”) NAV (ER-RTS) **316** to reflect the transaction duration indicated by the ER-RTS frame **310**. In response to the ER-RTS frame **310**, the non-ER destination **304** is configured to transmit a Clear-to-Send (CTS) frame **312**. The CTS frame **312** may include the address of the ER source **302** and a transaction duration based on the ER-RTS frame duration (e.g., such that it covers the reservation as per the ER-RTS duration). In this example, the ER OBSS device(s) **306** and non-ER OBSS device(s) **308** (e.g., devices in vicinity of the ER source **302**) detect the CTS frame and update respective NAV timers NAV (CTS) **318** and NAV (CTS) **320**.

[0053] Upon receiving the CTS frame **312** from the non-ER destination **304**, the ER source **302** may determine that the transmission channel is reserved and transmit an ER PPDU to the non-ER destination **304** in ‘n+1’ fragments **314-0**, . . . , **314-*n***, where n is a positive integer. Based on the reception of each of the n+1 fragments of the ER-PPDU frame, the non-ER destination **304** may be configured to transmit ‘n+1’ acknowledgements (ACKs) **316-0**, . . . , **316-*n*** (also referred to as ACK**0**) for the n+1 fragments of the ER-PPDU frame. For example, based on receiving an ER-PPDU fragment 1, ACK1 is transmitted by the non-ER destination **304**. In the illustrated example, the ER OBSS device **306** is further configured to update NAV (ER-fragment i) **322** corresponding to channel reservation based on the duration indicated in ER-PPDU (fragment i), which may include an ACK duration. In addition, the ER OBSS device **306** and the non-ER OBSS device **308** are further configured to update a respective NAV (ACK i) **324** and **326** corresponding to channel reservation based on the duration indicated by each ACK i **316**.

[0054] In a further example, the ER source **302** may not receive the CTS frame **312** or an ACK **316**. In such a scenario, the ER source **302** may be configured to wait for a time duration (e.g., short inter-frame spacing (SIFS)*2+ACK/CTS+slot time) and then transmit a contention-free end (CF-END) frame or ER-CF-END frame to the non-ER destination **304** in order to release the transmission channel. In another example, the non-ER destination **304** may not receive an ER-PPDU fragment for a pre-defined time duration or may receive a CF-END frame from the ER source **302**. In such a scenario, the non-ER destination **304** may transmit a CF-END frame to the ER source **302**. In a further example, the ordering of the CF-END frames is reversed.

[0055] Frame collisions may result due to the conventional RTS/CTS mechanism's inability to guarantee a transmission channel reservation at longer ranges. For example, non-ER devices which cannot operate in the ER range may detect the channel as free even if ER transmissions are ongoing, which may result in frame collisions. By reserving the channel for extended range using the first channel reservation scheme **300** (or any of the following channel reservations schemes) to reserve the channel for extended range, the possibility of frame collisions caused by hidden nodes can be mitigated.

[0056] FIG. **4** illustrates a table **400** presenting a stepwise summarization of the first channel reservation scheme **300** in accordance with an embodiment of the present disclosure. More particularly, the table **400** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the first channel reservation scheme **300** described above with reference to FIG. **3**. Each row of the table corresponds to one step.

[0057] FIG. **5** illustrates a second channel reservation scheme **500** in accordance with an embodiment of the present disclosure. In this example, a non-extended range (non-ER) source **502** may be required to communicate with an ER destination **504**. To facilitate such communication between the non-ER source **502** and the ER destination **504**, a transmission channel between the two devices may be reserved. When the transmission channel is reserved for the non-ER source **502** and the ER destination **504**, other devices within range may perceive the transmission channel as unavailable for use.

[0058] In the second channel reservation scheme **500**, the non-ER source **502** transmits a Request-to-Send (RTS) frame **510** to ER destination **504**. The RTS frame **510** contains a source address, a destination address, and a transaction duration. The transaction duration may indicate, for example, a time required for completion of the all of the related transactions between the non-ER source **502** and the ER destination **504**. In the illustrated example, an ER OBSS device(s) **506** and the non-ER OBSS device(s) **508** detect the RTS frame **510** and update a respective NAV (RTS) **518** and **516** to reflect the transaction duration indicated by the RTS frame **510**. In response to the RTS frame **510**, the ER destination **504** is configured to transmit an extended range Clear-to-Send (ER-CTS) frame. The ER-CTS frame may include the address of the non-ER source **502** and a transaction duration based on the RTS frame duration (e.g., such that it covers the reservation as per the RTS duration). In this example, the non-ER OBSS device(s) **508** (e.g., devices in vicinity of the non-ER source **502**) detect the ER-CTS frame and updates NAV (ER-CTS) **520**.

[0059] Upon receiving the ER-CTS frame **512** from the ER destination **504**, the non-ER source **502** may determine that the transmission channel is reserved and transmit a PPDU to the ER destination **504** in 'n+1' fragments **514-0**, . . . , **514-*n***, where n is a positive integer. Based on the reception of each of the n+1 fragments of the PPDU frame, the ER destination **504** may be configured to transmit 'n+1' extended range acknowledgements (ER-ACKs) **516-0**, . . . , **516-*n*** (also referred to as ER-ACK0-ER-ACKN) for the n+1 fragments of the ER-PPDU frame. For example, based on receiving a PPDU fragment 1, ER-ACK1 is transmitted by the ER destination **504**. In the illustrated example, the ER OBSS device **506** and the non-ER OBSS **508** are further configured to update a respective NAV (fragment i) **522** and **524** corresponding to channel reservation based on the duration indicated in PPDU (fragment i), which may include an ER-ACK duration. In addition, the non-ER OBSS device **508** is further configured to update a NAV (ER-ACK i) **526** corresponding to channel reservation based on the duration indicated by each ER-ACK i **516**.

[0060] In a further example, the non-ER source **502** may not receive the ER-CTS frame **512** or an ER-ACK **516**. In such a scenario, the non-ER source **502** may be configured to wait for a time duration (e.g., short inter-frame spacing (SIFS)*2+ACK/CTS+slot time) and then transmit a contention-free end (CF-END) frame in order to release the transmission channel. In another example, the ER destination **504** may not receive a PPDU fragment for a pre-defined time duration or may receive a CF-END frame from the non-ER source **502**. In such a scenario, the ER destination **504** may transmit a CF-END/ER-CF-END frame to the non-ER source **502**. In a further example, the ordering of the CF-END frames is reversed.

[0061] FIG. **6** illustrates a table **600** presenting a stepwise summarization of the second channel reservation scheme **500** in accordance with an embodiment of the present disclosure. More particularly, the table **600** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the second channel reservation scheme **500** described above with reference to FIG. **5**. Each row of the table corresponds to one step.

[0062] FIG. **7** illustrates a third channel reservation scheme **700** in accordance with an embodiment of the present disclosure. In this example, an extended range (ER) source **702** may be required to communicate with a non-ER destination **704**. To facilitate such communication between the ER source **702** and the non-ER destination **704**, a transmission channel between the two devices may be reserved. When the transmission channel is reserved for the ER source **702** and the non-ER destination **704**, other devices within range may perceive the transmission channel as unavailable for use.

[0063] In the third channel reservation scheme **700**, the ER source transmits a Request-to-Send (RTS) frame **710** to non-ER destination **704**, followed by an extended range Request-to-Send (ER-RTS) frame **712**. Each of the RTS frame **710** and the ER-RTS frame **712** contains a source address, a destination address, and a transaction duration. The transaction duration may indicate, for example, a time required for completion of the all of the related transactions between the ER source **702** and the non-ER destination **704**. In the illustrated example, a non-ER OBSS device **708** detects the RTS frame **710** and updates a corresponding NAV (RTS) **724** to reflect the transaction duration indicated by the RTS frame **710**, and an ER OBSS device **706** detects the ER-RTS frame **712** and sets or updates a corresponding NAV (ER-RTS) **726** to reflect the transaction duration indicated by the ER-RTS frame **712**. In response to the ER-RTS frame **712**, the non-ER destination **704** is configured to transmit a Clear-to-Send (CTS) frame **714**. The CTS frame **714** may include the address of the ER source **702** and a transaction duration based on the ER-RTS frame duration (e.g., such that it covers the reservation as per the ER-RTS duration). In this example, the ER OBSS device **706** and non-ER OBSS device **708** (e.g., devices in vicinity of the ER source **702**) detect the CTS frame **714** and update respective NAV timers NAV (CTS) **730** and NAV (CTS) **728**.

[0064] Upon receiving the CTS frame **712** from the non-ER destination **704**, the ER source **702** may determine that the transmission channel is reserved and transmit an ER PPDU to the non-ER destination **704** in 'n+1' fragments **716-0**, . . . , **716**-*n*, where n is a positive integer. Based on the reception of each of the n+1 fragments of the ER-PPDU frame, the non-ER destination **704** may be configured to transmit 'n+1' acknowledgements (ACKs) **718-0**, . . . , **718**-*n* (also referred to as ACK**0**) for the n+1 fragments of the ER-PPDU frame. For example, based on receiving an ER-PPDU fragment 1, ACK1 is transmitted by the non-ER destination **704**. In the illustrated example, the ER OBSS device **706** is further configured to update NAV (ER-fragment i) **732** corresponding to channel reservation based on the duration indicated in ER-PPDU (fragment i), which may include an ACK duration. In addition, the ER OBSS device **706** and the non-ER OBSS device **708** are further configured to update a respective NAV (ACK i) **734** and **736** corresponding to channel reservation based on the duration indicated by each ACK i **718**. In this example, the ER source **702** is further configured to transmit a CTS frame **720** (e.g., a CTS-to-self frame) following receipt of an ACK. In response, the non-ER OBSS device **708** updates a NAV (CTS) **738**.

[0065] In a further example, the ER source **702** may not receive the CTS frame **714** or an ACK **718**. In such a scenario, the ER source **702** may be configured to wait for a time duration (e.g., short inter-frame spacing (SIFS)*2+ ACK/CTS+slot time) and then transmit a contention-free end (CF-END) frame or ER-CF-END frame to the non-ER destination **704** in order to release the transmission channel. In another example, the non-ER destination **704** may not receive an ER-PPDU fragment for a pre-defined time duration or may receive a CF-END frame from the ER source **702**. In such a scenario, the non-ER destination **704** may transmit a CF-END frame to the ER source **702**. In a further example, the ordering of the CF-END frames is reversed.

[0066] FIG. **8** illustrates a table **800** presenting a stepwise summarization of the third channel reservation scheme **700** in accordance with an embodiment of the present disclosure. More particularly, the table **800** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the first channel reservation scheme **700** described above with reference to FIG. **7**. Each row of the table corresponds to one step.

[0067] FIG. **9** illustrates a table **900** presenting a stepwise summarization of a variation of the third channel reservation scheme **700** in accordance with an embodiment of the present disclosure. More particularly, the table **900** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the third channel reservation scheme **700** described above with reference to FIG. **7**, with the exception that in this variation the ER source **702** does not transmit a CTS-to-self frame after transmitting an ER fragment. Each row of the table corresponds to one step.

[0068] FIG. **10** illustrates a fourth channel reservation scheme **1000** in accordance with an embodiment of the present disclosure. In this example, a non-extended range (ER) source **1002** may be required to communicate with an ER destination **1004**. To facilitate such communication between the non-ER source **1002** and the ER destination **1004**, a transmission channel between the two devices may be reserved. When the transmission channel is reserved for the non-ER source **1002** and the ER destination **1004**, other devices within range may perceive the transmission channel as unavailable for use.

[0069] In the fourth channel reservation scheme **1000**, the non-ER source **1002** transmits a Request-to-Send (RTS) frame **1010** to ER destination **1004**, the RTS frame **1010** containing a source address, a destination address, and a transaction duration. The transaction duration may indicate, for example, a time required for completion of the all of the related transactions between the non-ER source **1002** and the ER destination **1004**. In the illustrated example, an ER OBSS device **1006** and a non-ER OBSS device **1008** detect the RTS frame **1010** and update a corresponding NAV (RTS) **1022** and **1024** to reflect the transaction duration indicated by the RTS frame **1010**. In response to the RTS frame **1010**, the ER destination **1004** is configured to transmit a Clear-to-Send (CTS) frame **1012** followed by an ER-CTS frame **1014**. Each of the CTS frame **1012** and CTS frame **1014** may include the address of the non-ER source **1002** and a transaction duration based on the RTS frame duration (e.g., such that it covers the reservation as per the RTS duration). In this example, the ER OBSS device **1006** and non-ER OBSS device **1008** (e.g., devices in vicinity of the non-ER source **1002**) detect the CTS frame **1012** and update respective NAV timers NAV (CTS) **1026** and NAV (CTS) **1028**. In addition, the ER OBSS device **1006** detects the ER-CTS frame **1014** and updates NAV (ER-CTS) **1026** to reflect the transaction duration indicated by the ER-CTS frame **1014**.

[0070] Upon receiving the ER-CTS frame **1014** from the ER destination **1004**, the non-ER source **1002** may determine that the transmission channel is reserved and transmit a PPDU to the ER destination **1004** in 'n+1' fragments **1016-0**, . . . , **1016**-*n*, where n is a positive integer. Based on the reception of each of the n+1 fragments of the PPDU frame, the ER destination **1004** may be configured to transmit 'n+1' extended range acknowledgements (ER-ACKs) **1018-0**, . . . , **1018**-*n* (also referred to as ER-ACK0-ER-ACKN) for the n+1 fragments of the PPDU frame. For example, based on receiving an PPDU fragment 1, ACK1 is transmitted by the ER destination **1004**. In the illustrated example, the ER OBSS device **1006** is further configured to update NAV (fragment i) **1030** corresponding to channel reservation based on the duration indicated in PPDU (fragment i), which may include an ACK duration. In addition, the ER OBSS device **1006** is further configured to update a NAV (ACK i) **1036** corresponding to channel reservation based on the duration indicated by each ACK i **1018**. In this example, the ER destination **1004** is further configured to transmit a CTS frame **1020** (e.g., a CTS-to-self frame) following receipt of an ACK. In response, the non-ER OBSS device **1008** updates a NAV (CTS) **1034**.

[0071] In a further example, the non-ER source **1002** may not receive the CTS frame **1012** or an ER-ACK **1018**. In such a scenario, the non-ER source **1002** may be configured to wait for a time duration (e.g., short inter-frame spacing (SIFS)*2+ACK/CTS+slot time) and then transmit a contention-free end (CF-END) frame to the ER destination **1004** in order to release the transmission channel. In another example, the ER destination **1004** may not receive an PPDU fragment for a pre-defined time duration or may receive a CF-END frame from the non-ER source **1002**. In such a scenario, the ER destination **1004** may transmit a CF-END frame to the non-ER source **1002**. In a further example, the ordering of the CF-END frames is reversed.

[0072] FIG. **11** illustrates a table **1100** presenting a step-wise summarization of the fourth channel reservation scheme **1000** in accordance with an embodiment of the present disclosure. More particularly, the table **1100** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the fourth channel reservation scheme **1000** described above with reference to FIG. **10**. Each row of the table corresponds to one step.

[0073] FIG. **12** illustrates a table **1200** presenting a step-wise summarization of a variation of the fourth channel reservation scheme **1000** in accordance with an embodiment of the present disclosure. More particularly, the table **1200** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the fourth channel reservation scheme **1000** described above with reference to FIG. **10**, with the exception that in this variation the ER destination **1004** does not transmit a CTS-to-self frame after transmitting an ER-ACK. Each row of the table corresponds to one step.

[0074] FIG. **13** illustrates a fifth channel reservation scheme **1300** in accordance with an embodiment of the present disclosure. In this example, an extended range (ER) source **1302** may be required to communicate with a non-ER destination **1304**. To facilitate such communication between the ER source **1302** and the non-ER destination **1304**, a transmission channel between the two devices may be reserved. When the transmission channel is reserved for the ER source **1302** and the non-ER destination **1304**, other devices within range may perceive the transmission channel as unavailable for use.

[0075] In the fifth channel reservation scheme **1300**, the ER source transmits an extended range Request-to-Send (ER-RTS) frame **1310** to non-ER destination **1304**, followed by a Request-to-Send (RTS) frame **1312**. Each of the RTS frame **1310** and the ER-RTS frame **1312** contains a source address, a destination address, and a transaction duration. The transaction duration may indicate, for example, a time required for completion of the all of the related transactions between the ER source **1302** and the non-ER destination **1304**. In the illustrated example, a non-ER OBSS device **1308** detects the RTS frame **1312** and updates a corresponding NAV (RTS) **1324** to reflect the transaction duration indicated by the RTS frame **1312**, and an ER OBSS device **1306** detects the ER-RTS frame **1310** and sets or updates a corresponding NAV (ER-RTS) **1326** to reflect the transaction duration indicated by the ER-RTS frame **1310**. In response to the RTS frame **1312**, the non-ER destination **1304** is configured to transmit a Clear-to-Send (CTS) frame **1314**. The CTS frame **1314** may include the address of the ER source **1302** and a transaction duration based on the RTS frame duration (e.g., such that it covers the reservation as per the RTS duration). In this example, the ER OBSS device **1306** and non-ER OBSS device **1308** (e.g., devices in vicinity of the ER source **1302**) detect the CTS frame **1314** and update respective NAV timers NAV (CTS) **1330** and NAV (CTS) **1328**.

[0076] Upon receiving the CTS frame **1312** from the non-ER destination **1304**, the ER source **1302** may determine that the transmission channel is reserved and transmit an ER PPDU to the non-ER destination **1304** in 'n+1' fragments **1316-0**, . . . , **1316-*n***, where n is a positive integer. Based on the reception of each of the n+1 fragments of the ER-PPDU frame, the non-ER destination **1304** may be configured to transmit 'n+1' acknowledgements (ACKs) **1318-0**, . . . , **1318-*n*** (also referred to as ACK0) for the n+1 fragments of the ER-PPDU frame. For example, based on receiving an ER-PPDU fragment 1, ACK1 is transmitted by the non-ER destination **1304**. In the illustrated example, the ER OBSS device **1306** is further configured to update NAV (ER-fragment i) **1332** corresponding to channel reservation based on the duration indicated in ER-PPDU (fragment i), which may include an ACK duration. In addition, the ER OBSS device **1306** and the non-ER OBSS device **1308** are further configured to update a respective NAV (ACK i) **1334** and NAV (ACK i) **1336** corresponding to channel reservation based on the duration indicated by each ACK i **1318**. In this example, the ER source **1302** is further configured to transmit a CTS frame **1320** (e.g., a CTS-to-self frame) following receipt of an ACK. In response, the non-ER OBSS device **1308** updates a NAV (CTS) **1338**.

[0077] In a further example, the ER source **1302** may not receive the CTS frame **1314** or an ACK **1318**. In such a scenario, the ER source **1302** may be configured to wait for a time duration (e.g., short inter-frame spacing (SIFS)*2+ ACK/CTS+slot time) and then transmit a contention-free end (CF-END) frame or ER-CF-END frame to the non-ER destination **1304** in order to release the transmission channel. In another example, the non-ER destination **1304** may not receive an ER-PPDU fragment for a pre-defined time duration or may receive a CF-END frame from the ER source **1302**. In such a scenario, the non-ER destination **1304** may transmit a CF-END frame to the ER source **1302**. In a further example, the ordering of the CF-END frames is reversed.

[0078] FIG. **14** illustrates a table **1400** presenting a step-wise summarization of the fifth channel reservation scheme **1300** in accordance with an embodiment of the present disclosure. More particularly, the table **1400** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the fifth channel reservation scheme **1300** described above with reference to FIG. **13**. Each row of the table corresponds to one step.

[0079] FIG. **15** illustrates a table **1500** presenting a step-wise summarization of a variation of the fifth channel reservation scheme **1300** in accordance with an embodiment of the present disclosure. More particularly, the table **1500** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the fifth channel reservation scheme **1300** described above with reference to FIG. **13**, with the exception that in this variation the ER source **1302** does not transmit a CTS-to-self frame after transmitting an ER fragment. Each row of the table corresponds to one step.

[0080] FIG. **16** illustrates a sixth channel reservation scheme **1600** in accordance with an embodiment of the present disclosure. In this example, an extended range (ER) source **1602** may be required to communicate with a non-ER destination **1604**. To facilitate such communication between the ER source **1602** and the non-ER destination **1604**, a transmission channel between the two devices may be reserved. When the transmission channel is reserved for the ER source **1602** and the non-ER destination **1604**, other devices within range may perceive the transmission channel as unavailable for use.

[0081] In the sixth channel reservation scheme **1600**, the ER source transmits an extended range Request-to-Send (ER-RTS) frame **1610** to non-ER destination **1604**, and then utilizes simultaneous CTS frames for extending the channel reservation in the non-ER OBSS device(s) **1608**. In particular, the ER source transmits a CTS-to-self frame **1612** at the same time the non-ER destination **1604** transmits a CTS frame **1614**. Each of the RTS frame **1610**, CTS-to-self frame **1612** and the CTS frame **1614** contains a source address, a destination address, and a transaction duration. The transaction duration may indicate, for example, a time required for completion of the all of the remaining transactions between the ER source **1602** and the non-ER destination **1604**. In the illustrated example, a non-ER OBSS device **1608** detects the ER-RTS frame **1610** and updates a corresponding NAV (ER-CTS) **1624** to reflect the transaction duration indicated by the ER-RTS frame **1610**, and an ER OBSS device **1606** detects the ER-RTS frame **1610** and sets or updates a corresponding NAV (ER-RTS) **1626** to reflect the transaction duration indicated by the ER-RTS frame **1610**. The CTS frame **1614** may include the address of the ER source **1602** and a transaction duration based on the RTS frame duration (e.g., such that it covers the reservation as per the RTS duration). In this example, the ER OBSS device **1606** and non-ER OBSS device **1608** (e.g., devices in vicinity of the ER source **1602**) detect the CTS-to-self frame **1612** and/or CTS frame **1614** and update respective NAV timers NAV (CTS) **1630** and NAV (CTS) **1628**.

[0082] Upon receiving the CTS frame **1612** from the non-ER destination **1604**, the ER source **1602** may determine that the transmission channel is reserved and transmit an ER PPDU to the non-ER destination **1604** in 'n+1' fragments **1616-0**, . . . , **1616-*n***, where n is a positive integer. Based on the reception of each of the n+1 fragments of the ER-PPDU frame, the non-ER destination **1604** may be configured to transmit 'n+1' acknowledgements (ACKs) **1618-0**, . . . , **1618-*n*** (also referred to as ACK0) for the n+1 fragments of the ER-PPDU frame. For example, based on receiving an ER-PPDU fragment 1, ACK1 is transmitted by the non-ER destination **1604**. In the illustrated example, the ER OBSS device **1606** is further configured to update NAV (ER-fragment i) **1632** corresponding to channel reservation based on the duration indicated in ER-PPDU (fragment i), which may include an ACK duration. In addition, the ER OBSS device **1606** and the non-ER OBSS device **1608** are further configured to update a respective NAV (ACK i) **1634** and NAV (ACK i) **1636** corresponding to channel reservation based on the duration indicated by each ACK i **1618**. In this example, the ER source **1602** is further configured to transmit a CTS frame **1620** (e.g., a CTS-to-self frame) following receipt of an ACK. In response, the non-ER OBSS device **1608** updates a NAV (CTS) **1638**.

[0083] In a further example, the ER source **1602** may not receive the CTS frame **1614** or an ACK **1618**. In such a scenario, the ER source **1602** may be configured to wait for a time duration (e.g., short inter-frame spacing (SIFS)*2+ ACK/CTS+slot time) and then transmit a contention-free end (CF-END) frame or ER-CF-END frame to the non-ER destination **1604** in order to release the transmission channel. In another example, the non-ER destination **1604** may not receive an ER-PPDU fragment for a pre-defined time duration or may receive a CF-END frame from the ER source **1602**. In such a scenario, the non-ER destination **1604** may transmit a CF-END frame to the ER source **1602**. In a further example, the ordering of the CF-END frames is reversed.

[0084] FIG. **17** illustrates a table **1700** presenting a step-wise summarization of the sixth channel reservation scheme in accordance with an embodiment of the present disclosure. More particularly, the table **1700** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the sixth channel reservation scheme **1600** described above with reference to FIG. **16**. Each row of the table corresponds to one step.

[0085] FIG. **18** illustrates a table **1800** presenting a step-wise summarization of a variation of the sixth channel reservation scheme **1600** in accordance with an embodiment of the present disclosure. More particularly, the table **1800** illustrates: (1) a transmitted packet/frame and the device transmitting the packet (e.g., ER source, non-ER source, ER destination or non-ER destination); (2) the function of the transmitted packet (e.g., channel reservations); and (3) a channel reservation duration based on each packet for the sixth channel reservation scheme **1600** described above with reference to FIG. **16**, with the exception that in this variation the ER source **1602** does not transmit a CTS-to-self frame after transmitting an ER fragment. Each row of the table corresponds to one step.

[0086] The scope of the present disclosure is not limited to a single ER destination or a single non-ER destination. In the various channel reservation schemes described herein, multiple ER destinations may be present. In such scenarios, simultaneous CTS frames or ER-CTS frames are transmitted to/from all ER destinations (e.g., when a source transmits a (extended) multi-user RTS (MU-RTS) frame. In an example, a when CTS frame is transmitted individually, either all ER destinations transmit a CTS/ER-CTS frame simultaneously or each ER destination transmits CTS/ER-CTS frames sequentially with each CTS/ER-CTS frame separated by a SIFS duration. When the ER source receives at least one

ER-CTS frame for ER transmission reservation or a CTS frame for non-ER transmission, the ER source may not send the CF-end frame and ER-CF-end frame and continue the ER transmission. The CF-end frame and the ER-CF-end frame may be sent only when none of the ER-CTS/CTS frames are received.

[0087] For simultaneous transmission of CTS or ER-CTS frames (e.g., in response to a MU-RTS frame), frame content may be the same and the destination may perform pre-compensation for carrier frequency offset and symbol clock error while transmitting the simultaneous CTS or ER-CTS frames. In some examples, the pre-compensation uses the value from the previously received ER-RTS frame, and if no ER-RTS frame was transmitted from the source then the pre-compensation uses the values from a previously received RTS frame.

[0088] FIG. **19** is a flow diagram illustrating an example method **1900** for channel reservation in accordance with an embodiment of the present disclosure. The method **1900** can be performed, for example, by the AP **102** or STA **116** described with reference to FIG. **1**. The method **1900** may be utilized, for example, by an ER source to perform channel reservation operations such as described with reference to any of FIG. **3**, FIG. **7**, FIG. **13**, or FIG. **16**.

[0089] The method begins at step **1902** where an ER source transmits an extended range (ER) Request to Send (RTS) frame (e.g., for reception by a non-ER wireless device/destination) over a transmission channel. The method continues at step **1904**, where the ER source receives a non-ER Clear to Send (CTS) frame in response to the ER RTS frame. Following receipt of the non-ER CTS frame, the method continues at step **1906** where the ER source transmits an ER PPDU (or a fragment of the ER PPDU) to the non-ER wireless device. In response, the ER source receives (at step **1908**) a non-ER ACK from the non-ER wireless device. In this manner, OBSS wireless devices (e.g., ER OBSS devices and/or non-ER devices) within range of the illustrated frame exchange (or portions thereof) may update their respective NAV timers to indicate that the transmission channel is busy during the transmission of the ER PPDU, thereby mitigating the possibility of potential frame collisions in the transmission channel.

[0090] FIG. **20** is a flow diagram illustrating another example method **2000** for channel reservation in accordance with an embodiment of the present disclosure. The method **2000** can be performed, for example, by the AP **102** or STA **116** described with reference to FIG. **1**. The method **2000** may be utilized, for example, by a non-ER source to perform channel reservation operations such as described with reference to FIG. **5** and FIG. **10**.

[0091] The method begins at step **2002** where a non-ER source transmits a non-ER RTS frame (e.g., for reception by an ER wireless device/destination) over a transmission channel. The method continues at step **2004**, where the non-ER source receives an ER CTS frame in response to the non-ER RTS frame. Following receipt of the ER CTS frame, the method continues at step **2006** where the non-ER source transmits a non-ER PPDU (or a fragment of the non-ER PPDU) to the ER wireless device. In response, the non-ER source receives (at step **2008**) an ER ACK from the ER wireless device. In this manner, OBSS wireless devices (e.g., ER OBSS devices and/or non-ER devices) within range of the illustrated frame exchange (or portions thereof) may update their respective NAV timers to indicate that the transmission channel is busy during the transmission of the non-ER PPDU, thereby mitigating the possibility of potential frame collisions in the transmission channel.

[0092] FIG. **21** illustrates an example of an Extended Range (ER)/Enhanced Long Range (ELR) physical layer protocol data unit (PPDU) **2100** that may be utilized in the examples described herein. The ER/ELR PPDU of this example includes a legacy preamble **2102**, an ER/ELR preamble **2104**, and ER/ELR data **2106**. In an example, the ER/ELR PPDU **2100** is legacy ER/ELR PPDU. In another example, such as described in greater detail with reference to FIG. **22**, the ER/ELR PPDU **2100** is non-legacy ELR PPDU.

[0093] FIG. **22** illustrates an example of an Enhanced Long Range (ELR) physical layer protocol data unit (PPDU) **2200**. The ELR PPDU **2200** of this example includes a legacy portion **2202** and an ELR portion **2216**, which are transmitted as a waveform. The legacy portion **2202** includes legacy fields which legacy 802.11 devices are able to decode for co-existence while the ELR portion **2216** may include one or more ELR fields so that next generation devices (e.g., Wi-Fi 8 UHR devices) are able to transmit and receive data in the ELR portion **2216** with increased range and lower SNR. In an example, a bandwidth of the legacy portion **2202** and the ELR portion **2216** is the same to provide co-existence with legacy devices.

[0094] The legacy portion **2202** of this example includes a legacy short training field (L-STF) **2204**, a legacy long training field (L-LTF) **2206**, a legacy signal (L-SIG) field **2208**, a repeated L-SIG (RL-SIG) field **2210**, and a universal signaling (U-SIG) field **2212**. The L-STF **2204** is used by a recipient device to detect the start of the PPDU or portion thereof and to establish orthogonal frequency division multiplexed/access (OFDM/A) symbol timing for data detection, i.e. frame acquisition and time synchronization. The L-LTF **2206** is used for channel estimation/training for information detection. Channel estimation is a process of determining channel characteristics (e.g., a frequency response) of a channel in which the PPDU is transmitted. The L-SIG field **2208** includes information for data decoding and coexistence such as a 12 bit packet length value (LENGTH), rate information, etc. In an example, LENGTH is signaled to spoof legacy devices for purposes of clear channel assessment (CCA), and non-legacy devices can decode a TXOP for CCA. In addition, a non-legacy device (e.g., an intended receiver) may also derive a Nsym (with may also be referred to as Length) value from the L_SIG field **2208**. However, as L-SIG LENGTH decoding may not be reliable, this information may be repeated in the ELR-SIG field **2222**. In an example, a number of data symbols (Nsym/Length) subfield can be directly signaled in the ELR-SIG field **2222** utilizing fewer than 12 bits (e.g., 8 or 9 bits), thereby saving 3-4 bits of signaling and simplifying packet length calculations by receiving devices.

[0095] In an example, the L-SIG **2208** may be repeated in time and the repetition is included in the repeated RL-SIG field **2210** of the legacy portion **2202** such that the L-SIG **2208** is repeated twice. The repetition may allow increased range and SNR associated with receipt of the L-SIG field **2208**. The U-SIG field **2212** in the legacy portion **2202** may include an indication of a version of the physical layer communication of IEEE 802.11 such as in a three-bit PHY identifier, an uplink/downlink flag, Basic Service Set (BSS) color, transmission (TX) opportunity (TXOP) duration, bandwidth, etc. Like the L-SIG field, the U-SIG field **2212** may also be repeated twice in time for better reception. In the illustrated ELR PPDU, the U-SIG field **2212** includes a U-SIG-**1** subfield **2226** and a U-SIG-**2** subfield **2228**. To also extend the range, a transmission power of a waveform of one or more of the L-STF **2204** and the L-LTF **2206** may be boosted to 3 dB.

[0096] To further extended the range, a transmission power associated with the L-STF **2204** and L-LTF **2206** could be boosted to greater than 3 dB and the L-SIG field **2208** and the U-SIG field **2212** may be repeated more than twice, but these changes may create compatibility issues with legacy devices because of energy drop-off between when the legacy device receives the L-SIG **2208** and receives the L-LTF **2206**, erroneous detection of the L-SIG field **2208** and U-SIG field **2212**, and a high peak to average power ratio of the PPDU. The legacy portion **2202** may be modulated on an orthogonal frequency division multiplexed (OFDM) signal which defines subcarriers for transmitting the fields of the legacy portion **2202** and as a result range extension is also limited by a maximum peak to average ratio (PAPR) of the waveform representing the PPDU which IEEE 802.11 specifies. IEEE 802.11b defines a single-carrier binary sequence design which demonstrates range extension benefits over OFDM associated with 802.11ax and 802.11bc. However, the carrier is only defined for a 2.4 GHz band and does not co-exist with IEEE 802.11a such that the format cannot be extended into a 5 GHZ and 6 GHz band without also causing backward compatibility issues for legacy devices.

[0097] In some examples, one or more transition symbols may be optionally added after the U-SIG field **2212** in the legacy portion **2202** preceding the ELR portion **2216**. In the illustrated example, an ELR-MARK field **2214** is included. The ELR-MARK field **2214** may be a symbol, such as an OFDM symbol, which spans a channel bandwidth and has a predefined duration, and may signal a transition between the U-SIG field **2212** and the ELR portion **2216**. In an example, a non-legacy wireless device receiving the ELR PPDU **2200** may need to determine a receiver state machine based on a U-SIG decoding CRC check. In the event that the U-SIG decoding fails, e.g., a CRC check does not pass, the wireless device needs to reset receive time domain parameters, such as CFO and sample frequency offset (SFO) compensation, while ELR preamble detection logic is still running. The ELR-MARK field **2214** may provide some buffer time such that the ELR preamble will not arrive before the receive time domain parameters is reset. Thus, the ELR preamble detection will not be affected by the status of the legacy preamble detection. In one example, the ELR-MARK field **2214** is defined as a signaling field (with predefined tone patterns), similar to EHT-SIG as in IEEE 802.11bc. In another example, the ELR-MARK field **2214** is a predefined sequence, which can further include a BSS color indication (e.g., a value of 0 to 63) for use by receiving devices to determine if the received PPDU is an ELR PPDU and if the ELR PPDU is from OBSS.

[0098] To achieve range extension, the legacy portion **2202** is followed by the ELR portion **2216**. By appending the legacy portion **2202** to the ELR portion **2216**, the ELR PPDU **2200** is able to co-exist with the 802.11 legacy devices. In an example, a PPDU length in octets indicated in the L-SIG field **2208** is backward compatible with legacy devices to detect the ELR PPDU **2200** while the U-SIG field **2212** provides both backward and forward compatibility. For example, the U-SIG field **2212** is modulated with binary phase shift keying (BPSK), and the U-SIG field **2212** may indicate a "PHY version identifier" which indicates a PHY version. To signal the new ELR format, a new value of "PHY version identifier" in the U-SIG field **2212** can be used to indicate next generation PHY and a new "PPDU format" subfield can indicate the new ELR format. In an example, the ELR PPDU **2200** may be limited for transmission over one spatial stream using modulation coding scheme (MCS) 0 or lower data rates. In an example, an unintended receiver which does not support a non-legacy standard can use indications of these fields to enter into a power save state when the ELR PPDU **2200** is received and is not able to be processed, and set network allocation vector (NAV) values correspondingly to not transmit for at least a PPDU duration.

[0099] The ELR portion **2216** of the illustrated example includes an ELR preamble and a UHR-Data field **2224**. The ELR preamble includes a UHR short training field (UHR-STF) **2218**, a UHR long training field (UHR-LTF) **2220**, and an ELR signal (ELR-SIG) field **2222**. The UHR-STF **2218** may be a predefined binary sequence used to detect the start of the ELR portion **2216** and provide symbol timing for data detection, i.e. frame acquisition and time synchronization. In one embodiment, the UHR-STF **2218** consists of two parts: one binary sequence for synchronization followed by one binary sequence for STF ending and UHR-LTF **2220** may not be included. In another embodiment, the UHR-STF **2218** consists of one binary sequence followed by UHR-LTF **2220**. If the receiver is not able to detect the legacy STF **2204**, the receiver will attempt to detect the UHR-STF **2218**. The UHR-LTF **2220** defines a binary sequence for channel estimation/training by a receiver. In some examples, this field may be omitted for certain modulation schemes such as differential encoding for 802.11b.

[0100] The ELR-SIG field **2222** includes information for data decoding. The ELR-SIG field **2222** may include various parameters including a modulation and coding scheme (MCS) subfield, a coding subfield that indicates whether BCC or LDPC is used, a number of symbols (Nsym) or Length subfield that indicates a number of ELR data symbols, a cyclic redundancy check (CRC), etc., defined by an ELR-SIG binary sequence. In an example, the ELR-SIG field **2222** includes two symbols (i.e., an ELR-SIG-1 subfield **2230** and an ELR-SIG-2 subfield **2232**). Forward error correction (FEC) coding may be defined for the ELR-SIG field **2222** to enhance reliability, e.g. binary convolutional coding (BCC). The UHR-Data field **2224** which follows the ELR preamble includes a data payload defined by an ELR-data binary sequence. Forward error correction (FEC) coding may be defined to enhance data decoding reliability, e.g. BCC or low density parity check code (LDPC).

[0101] The ELR portion **2216** may be transmitted in various ways. In one example, a waveform representative of the binary sequences of the ELR portion **2216** may be defined with a low peak-to-average ratio (PAPR) such that the transmitter can increase the maximum transmit power to increase communication range or enhance receiver reception reliability. Because the legacy preamble **2202** may already have a high PAPR, a power amplifier associated with the transceiver which transmits the ELR portion **2216** may back off by ~10 dB to keep all samples which are to be transmitted in a linear region to accommodate the PAPR. The power amplifier may transmit the ELR portion **2216** with some peak samples into a non-linear region for range extension and an ER spectrum growth due to the non-linearity may result in a lower PAPR, close to 0 dB depending on binary sequence design. In an example, the ELR portion **2216** may be transmitted with a power similar to a peak power of the legacy portion **2202** with 10 dB gain, but in some cases, an increase in transmit power may be limited by a power spectral density. In another example, the transmit power of a waveform of the ELR portion **2216** may be set to a power boost such as 3 dB or the transmitter may set a power boost based on a historical transmit power range.

[0102] The binary sequence of the UHR-STF **2218** may be modulated on a time domain waveform. Time domain modulation is defined as varying a modulation of a waveform over time. The binary sequence of the UHR-LTF **2220**, ELR-SIG **2222**, and UHR-Data field **2224** may be transmitted based on single carrier (SC) time-domain multiplexing (TDM). A binary sequence may be directly modulated on a time domain waveform to generate different time domain signals for different binary sequences and additional spreading can be applied, e.g. 802.11b direct sequence spread spectrum (DSSS).

[0103] In another example, the modulation of one or more of the fields in the ELR preamble may be based on a single carrier (SC) frequency-domain multiplexing (FDM). Frequency domain multiplexing is defined as loading binary sequences to be transmitted onto subcarriers in a frequency band versus time domain signals, where different frequency bands may be assigned to different wireless devices. The UHR-STF **2218** may be transmitted with one of the 802.11b DSSS, a zero correlation zone (ZCZ) spreading sequence, or a Golay sequence (defined in 802.11ad/ay). The UHR-LTF **2220** may include a predefined binary sequence to estimate a channel of each subcarrier and transmitted in a manner similar to the UHR-STF **2218**. The ELR-SIG field **2222** and the UHR-Data field **2224** may be transmitted with SC-FDM. An LTF1 subfield of UHR-LTF **2220** may be added before the ELR-SIG field **2222** to indicate information to demodulate SIG content and an LTF2 subfield of the UHR-LTF **2220** may be added to indicate information to demodulate UHR-Data field **2224** content. The information may indicate a tone mapping and the LTF2 may be included in the UHR-LTF **2220** when a tone mapping of the subcarriers on which a binary sequence of the information are loaded and/or a bandwidth of the UHR-Data field **2224** is different from the ELR-SIG field **2222**. The tone mapping may be a process of selecting subcarriers in a set of subcarriers to transmit the binary sequence, where a subcarrier or tone is a defined frequency or frequencies in a channel bandwidth such as a 20 MHz channel having an amplitude and a phase. In an example, a bit or bits of the sequence may be modulated on the tone such as by binary phase shift keying (BPSK) or quadrature phase shift keying (QPSK) to form a waveform.

[0104] In the example of FIG. **22**, the ELR preamble may have a fixed transmission format as the ELR-SIG field **2222** follows the UHR-STF **2218** and UHR-LTF **2220**. For example, the UHR-LTF **2220** transmission format may be fixed at 4xLTF with 4 repetitions, 1xLTF with 16 repetitions, 4xLTF with sparse tone loading and 2 repetitions, etc.

[0105] While the innovate aspects of the present disclosure have been generally described in the context of the 802.11bn amendment (and future generations) of the IEEE 802.11 standard, a person having ordinary skill in the art will readily recognize that teachings and concepts herein may be applied to other wireless networks and standards including, for example, Long Term Evolution (LTE) standards and Bluetooth standards.

[0106] The innovative channel reservations schemes illustrated in the drawings and described herein minimize potential frame collisions and interference in wireless networks involving devices having disparate transmission ranges. In an illustrative, non-limiting embodiment, a method is provided for reserving a transmission channel by a first wireless device. The method includes transmitting an extended range Request to Send (RTS) frame and receiving, in response, a non-extended range Clear to Send (CTS) frame from a second wireless device. The method further includes transmitting an extended range physical layer protocol data unit (PPDU) for reception by the second wireless device.

[0107] The method of this embodiment includes optional aspects. With one optional aspect, transmitting the extended range PPDU includes fragmenting the extended range PPDU into a plurality of extended range fragments and transmitting a first extended range fragment of the plurality of extended range fragments. This optional aspect further includes receiving a non-extended range acknowledgement (ACK) from the second wireless device and transmitting a second extended range fragment of the plurality of extended range fragments. In another optional aspect, in response to detecting a failure to receive at least one of the non-extended range CTS frame or the non-extended range ACK from the second wireless device, the first wireless device transmits a contention-free end (CF-END) frame for reception by the second wireless device. In yet another optional aspect, the first wireless device transmits a non-extended range RTS frame prior to transmitting the second extended range fragment. In yet another optional aspect, the first wireless device transmits a non-extended range CTS frame (e.g., a CTS-to-self frame) prior to transmitting the second extended range fragment.

[0108] In another optional aspect, transmitting the extended range PPDU for reception by the second wireless device includes transmitting the extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device. In this optional aspect, the method further includes sequentially receiving a non-extended range acknowledgement (ACK) from at least two of the plurality of wireless devices (separated by a Short Interframe Spacing (SIFS)) in response to transmitting the extended range PPDU, and further transmitting a second extended range PPDU for reception by the plurality of wireless devices. In yet another optional aspect, the extended range RTS frame is a multi-user RTS (MU-RTS) frame and transmitting the extended range PPDU further includes transmitting the extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device. This optional aspect further includes simultaneously receiving a non-extended range acknowledgement (ACK) from at least two of the plurality of wireless devices and transmitting a second extended range PPDU for reception by the plurality of wireless devices.

[0109] In another illustrative, non-limiting embodiment, a method is provided for reserving a transmission channel by a first wireless device. The method of this embodiment includes transmitting a non-extended range Request to Send (RTS) frame and receiving, in response, an extended range Clear to Send (CTS) frame from a second wireless device.

The method further includes transmitting a non-extended range physical layer protocol data unit (PPDU) for reception by the second wireless device.

[0110] The method of this embodiment includes optional aspects. With one optional aspect, transmitting the non-extended range PPDU includes fragmenting the non-extended range PPDU into a plurality of extended range fragments and transmitting a first non-extended range fragment of the plurality of extended range fragments. This optional aspect further includes receiving an extended range acknowledgement (ACK) from the second wireless device and transmitting a second non-extended range fragment of the plurality of extended range fragments. In another optional aspect, in response to detecting a failure to receive at least one of the extended range CTS frame or the extended range ACK from the second wireless device, the first wireless device transmits a contention-free end (CF-END) frame for reception by the second wireless device. In yet another optional aspect, the first wireless device receives a non-extended range CTS frame from the second wireless device prior to transmitting the non-extended range PPDU.

[0111] In another optional aspect, transmitting the non-extended range PPDU for reception by the second wireless device includes transmitting the non-extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device. In this optional aspect, the method further includes sequentially receiving an extended range acknowledgement (ACK) from at least two of the plurality of wireless devices (separated by a Short Interframe Spacing (SIFS)) in response to transmitting the non-extended range PPDU, and further transmitting a second non-extended range PPDU for reception by the plurality of wireless devices. In yet another optional aspect, the non-extended range RTS frame is a multi-user RTS (MU-RTS) frame and transmitting the non-extended range PPDU further includes transmitting the non-extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device. This optional aspect further includes simultaneously receiving a non-extended range acknowledgement (ACK) from at least two of the plurality of wireless devices and transmitting a second non-extended range PPDU for reception by the plurality of wireless devices.

[0112] With another illustrative, non-limiting embodiment, a wireless device includes one or more wireless transceivers and one or more processors operably coupled to the one or more wireless transceivers. The one or more processors are arranged to execute the operational instructions to transmit an extended range Request to Send (RTS) frame and receive, in response, a non-extended range Clear to Send (CTS) frame from a second wireless device. The one or more processors of the wireless device are further arranged to transmit an extended range physical layer protocol data unit (PPDU) for reception by the second wireless device.

[0113] The embodiment includes optional aspects. With one optional aspect, transmitting the extended range PPDU includes fragmenting the extended range PPDU into a plurality of extended range fragments and transmitting a first extended range fragment of the plurality of extended range fragments. In this optional aspect, the one or more processors of the wireless device are further arranged to receive a non-extended range acknowledgement (ACK) from the second wireless device and transmit a second extended range fragment of the plurality of extended range fragments. In another optional aspect, the first wireless device transmits a non-extended range RTS frame prior to transmitting the second extended range fragment. In yet another optional aspect, the first wireless device transmits a non-extended range CTS frame (e.g., a CTS-to-self frame) prior to transmitting the second extended range fragment.

[0114] In another optional aspect, the extended range RTS frame is a multi-user RTS (MU-RTS) frame and transmitting the extended range PPDU for reception by the second wireless device further includes transmitting the extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device. In this optional aspect, the one or more processors are further arranged to simultaneously receive a non-extended range acknowledgement (ACK) from at least two of the plurality of wireless devices in response to transmitting the extended range PPDU, and transmit a second extended range PPDU for reception by the plurality of wireless devices.

[0115] In another illustrative, non-limiting embodiment, a wireless device includes one or more wireless transceivers and one or more processors operably coupled to the one or more wireless transceivers. The one or more processors are arranged to execute the operational instructions to transmit a non-extended range Request to Send (RTS) frame and receive, in response, an extended range Clear to Send (CTS) frame from a second wireless device. The one or more processors of the wireless device are further arranged to transmit a non-extended range physical layer protocol data unit (PPDU) for reception by the second wireless device.

[0116] The embodiment includes optional aspects. With one optional aspect, transmitting the non-extended range PPDU includes fragmenting the non-extended range PPDU into a plurality of non-extended range fragments and transmitting a first non-extended range fragment of the plurality of non-extended range fragments. In this optional aspect, the one or more processors of the wireless device are further arranged to receive an extended range acknowledgement (ACK) from the second wireless device and transmit a second non-extended range fragment of the plurality of non-extended range fragments. In another optional aspect, the one or more processors of the wireless device are further arranged to receive an extended range CTS frame from the second wireless device prior to transmitting the non-extended range PPDU.

[0117] To implement various operations described herein, computer program code (i.e., program instructions for carrying out these operations) may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, Smalltalk, Python, C++, or the like, conventional procedural programming languages, such as the "C" programming language or similar programming languages, or any of machine learning software. These program instructions may also be stored in a computer readable storage medium that can direct a computer system, other programmable data processing apparatus, controller, or other device to operate in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the operations specified in the block diagram block or blocks. The program instructions may also be loaded onto a processing core, processing circuitry, computer, other programmable data processing apparatus, controller, or other device to cause a series of operations to be performed on the computer, or other programmable apparatus or devices, to produce a computer implemented process such that the instructions upon execution provide processes for implementing the operations specified in the block diagram block or blocks.

[0118] As may be used herein, the term(s) “configured to”, “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via an intervening item (e.g., an item includes, but is not limited to, a component, an element, a circuit, and/or a module) where, for an example of indirect coupling, the intervening item does not modify the information of a signal but may adjust its current level, voltage level, and/or power level. As may further be used herein, inferred coupling (i.e., where one element is coupled to another element by inference) includes direct and indirect coupling between two items in the same manner as “coupled to”.

[0119] As may further be used herein, the term(s) “arranged to”, “configured to”, “operable to”, “coupled to”, or “operably coupled to” indicates that an item includes one or more of power connections, input(s), output(s), etc., to perform, when activated, one or more its corresponding functions and may further include inferred coupling to one or more other items. As may still further be used herein, the term “associated with” includes direct and/or indirect coupling of separate items and/or one item being embedded within another item.

[0120] As may be used herein, one or more claims may include, in a specific form of this generic form, the phrase “at least one of a, b, and c” or of this generic form “at least one of a, b, or c”, with more or less elements than “a”, “b”, and “c”. In either phrasing, the phrases are to be interpreted identically. In particular, “at least one of a, b, and c” is equivalent to “at least one of a, b, or c” and shall mean a, b, and/or c. As an example, it means: “a” only, “b” only, “c” only, “a” and “b”, “a” and “c”, “b” and “c”, and/or “a”, “b”, and “c”.

[0121] As may also be used herein, the terms “processor”, “processing circuitry”. “processing circuit”, “processing module”, and/or “processing unit” may be a single processing device or a plurality of processing devices. Such a processing device may be a microprocessor, microcontroller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions. Further, such a processing device may include a plurality of processing cores or processing domains, which may operate on separate power domains. The processor, processing circuitry, processing circuit, processing module, and/or processing unit may be (or may further include) memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processor, processing circuitry, processing circuit, processing module, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processor, processing circuitry, processing circuit, processing module, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network). Further note that if the processor, processing circuitry, processing circuit, processing module, and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processor, processing circuitry, processing circuit, processing module, and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the figures. Such a memory device or memory clement can be included in an article of manufacture.

[0122] One or more embodiments have been described above with the aid of method steps illustrating the performance of specified functions and relationships thereof. The boundaries and sequence of these functional building blocks and method steps have been arbitrarily defined herein for convenience of description. Alternate boundaries and sequences can be defined so long as the specified functions and relationships are appropriately performed. Any such alternate boundaries or sequences are thus within the scope and spirit of the claims.

[0123] To the extent used, the logic diagram block boundaries and sequence could have been defined otherwise and still perform the certain significant functionality. Such alternate definitions of both functional building blocks and logic diagram blocks and sequences are thus within the scope and spirit of the claims. One of average skill in the art will also recognize that the functional building blocks, and other illustrative blocks, modules and components herein, can be implemented as illustrated or by discrete components, application specific integrated circuits, processors/processing cores executing appropriate software and the like or any combination thereof.

[0124] The one or more embodiments are used herein to illustrate one or more aspects, one or more features, one or more concepts, and/or one or more examples. A physical embodiment of an apparatus, an article of manufacture, a machine, and/or of a process may include one or more of the aspects, features, concepts, examples, etc. described with reference to one or more of the embodiments discussed herein. Further, from figure to figure, the embodiments may incorporate the same or similarly named functions, steps, modules, etc. that may use the same or different reference numbers and, as such, the functions, steps, modules, etc. may be the same or similar functions, steps, modules, etc. or different ones.

[0125] The term “module” may be used in the description of one or more of the embodiments. A module implements one or more functions via a device such as a processor or other processing device or other hardware that may include or operate in association with a memory that stores operational instructions. A module may operate independently and/or in conjunction with software and/or firmware. As also used herein, a module may contain one or more submodules, each of which may be one or more modules.

[0126] As may further be used herein, a computer readable memory includes one or more memory elements. A memory element may be a separate memory device, multiple memory devices, or a set of memory locations within a memory device. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, a quantum register or other quantum memory and/or any other device that stores data in a non-transitory manner. Furthermore, the memory device may be in a form of a solid-state memory, a hard drive memory or other disk storage, cloud memory, thumb drive, server memory, computing device memory, and/or other non-transitory medium for storing data. The storage of data includes temporary storage (i.e., data is lost when power is removed from the memory element) and/or persistent storage (i.e., data is retained when power is removed from the memory element). As used herein, a transitory medium shall mean one or more of: (a) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for temporary storage or persistent storage; (b) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for temporary storage or persistent storage; (c) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for processing the data by the other computing device; and (d) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for processing the data by the other element of the computing device. As may be used herein, a non-transitory computer readable memory is substantially equivalent to a computer readable memory. A non-transitory computer readable memory can also be referred to as a non-transitory computer readable storage medium.

[0127] While particular combinations of various functions and features of the one or more embodiments have been expressly described herein, other combinations of these features and functions are likewise possible. The present disclosure is not limited by the particular examples disclosed herein and expressly incorporates these other combinations.

What is claimed is:

1. A method for reserving a transmission channel by a first wireless device, the method comprising:

transmitting an extended range Request to Send (RTS) frame;

receiving, in response to the extended range RTS frame, a non-extended range Clear to Send (CTS) frame from a second wireless device; and

transmitting an extended range physical layer protocol data unit (PPDU) for reception by the second wireless device.

2. The method of claim **1**, wherein transmitting the extended range PPDU for reception by the second wireless device includes:

fragmenting the extended range PPDU into a plurality of extended range fragments;

transmitting a first extended range fragment of the plurality of extended range fragments;

receiving a non-extended range acknowledgement (ACK) from the second wireless device; and

transmitting a second extended range fragment of the plurality of extended range fragments.

3. The method of claim **2**, further comprising:

in response to detecting a failure to receive at least one of the non-extended range CTS frame or the non-extended range ACK from the second wireless device, transmitting a contention-free end (CF-END) frame for reception by the second wireless device.

4. The method of claim **1**, further comprising:

prior to transmitting the extended range PPDU, transmitting a non-extended range RTS frame.

5. The method of claim **1**, further comprising:

prior to transmitting the extended range PPDU, transmitting a non-extended range CTS frame.

6. The method of claim **1**, wherein transmitting the extended range PPDU for reception by the second wireless device includes transmitting the extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device, the method further including:

in response to transmitting the extended range PPDU, sequentially receiving a non-extended range acknowledgement (ACK) from at least two of the plurality of wireless devices, wherein consecutive ACKs are separated by a Short Interframe Spacing (SIFS); and

transmitting a second extended range PPDU for reception by the plurality of wireless devices.

7. The method of claim **1**, wherein the extended range RTS frame is a multi-user RTS (MU-RTS) frame and wherein transmitting the extended range PPDU for reception by the second wireless device further includes transmitting the extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device, the method further including:

in response to transmitting the extended range PPDU, simultaneously receiving a non-extended range acknowledgement (ACK) from at least two of the plurality of wireless devices; and

transmitting a second extended range PPDU for reception by the plurality of wireless devices.

8. A method for reserving a transmission channel by a first wireless device, the method comprising:

transmitting a non-extended range Request to Send (RTS) frame;

receiving, in response to the non-extended range RTS frame, an extended range Clear to Send (CTS) frame from a second wireless device; and

transmitting a non-extended range physical layer protocol data unit (PPDU) for reception by the second wireless device.

9. The method of claim **8**, wherein transmitting the non-extended range PPDU for reception by the second wireless device includes:

fragmenting the non-extended range PPDU into a plurality of non-extended range fragments;

transmitting a first non-extended range fragment of the plurality of non-extended range fragments;

receiving an extended range acknowledgement (ACK) from the second wireless device; and

transmitting a second non-extended range fragment of the plurality of non-extended range fragments.

10. The method of claim **9**, further comprising:

in response to detecting a failure to receive at least one of the extended range CTS frame or the extended range ACK from the second wireless device, transmitting a contention-free end (CF-END) frame for reception by the second wireless device.

11. The method of claim **8**, further comprising:

prior to transmitting the non-extended range PPDU, receiving a non-extended range CTS frame from the second wireless device.

12. The method of claim **8**, wherein transmitting the non-extended range PPDU for reception by the second wireless device includes transmitting the non-extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device, the method further comprising:

in response to transmitting the non-extended range PPDU, sequentially receiving an extended range acknowledgement (ACK) from at least two of the plurality of wireless devices, wherein consecutive ACKs are separated by a Short Interframe Spacing (SIFS); and

transmitting a second non-extended range PPDU for reception by the plurality of wireless devices.

13. The method of claim **8**, wherein the non-extended range RTS frame is a multi-user RTS (MU-RTS) frame and wherein transmitting the non-extended range PPDU for reception by the second wireless device further includes transmitting the non-extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device, the method further comprising:

in response to transmitting the extended range PPDU, simultaneously receiving an extended range acknowledgement (ACK) from at least two of the plurality of wireless devices; and

transmitting a second non-extended range PPDU for reception by the plurality of wireless devices.

14. A wireless device, comprising:

one or more wireless transceivers; and

one or more processors operably coupled to the one or more wireless transceivers, wherein the one or more processors are arranged to:

transmit an extended range Request to Send (RTS) frame;

receive, in response to the extended range RTS frame, a non-extended range Clear to Send (CTS) frame from a second wireless device; and

transmit an extended range physical layer protocol data unit (PPDU) for reception by the second wireless device.

15. The wireless device of claim **14**, wherein transmitting the extended range PPDU for reception by the second wireless device includes:

fragmenting the extended range PPDU into a plurality of extended range fragments;

transmitting a first extended range fragment of the plurality of extended range fragments;

receiving a non-extended range acknowledgement (ACK) from the second wireless device; and

transmitting a second extended range fragment of the plurality of extended range fragments.

16. The wireless device of claim **14**, wherein the one or more processors are further arranged to:

prior to transmitting the extended range PPDU, transmit a non-extended range RTS frame.

17. The wireless device of claim **14**, wherein the one or more processors are further arranged to:

prior to transmitting the extended range PPDU, transmit a non-extended range CTS frame.

18. The wireless device of claim **14**, wherein transmitting the extended range PPDU for reception by the second wireless device includes transmitting the extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device, and wherein the one or more processors are further arranged to:

in response to transmitting the extended range PPDU, sequentially receive a non-extended range acknowledgement (ACK) from at least two of the plurality of wireless devices, wherein consecutive ACKs are separated by a Short Interframe Spacing (SIFS); and

transmit a second extended range PPDU for reception by the plurality of wireless devices.

19. The wireless device of claim **14**, wherein the extended range RTS frame is a multi-user RTS (MU-RTS) frame and wherein transmitting the extended range PPDU for reception by the second wireless device further includes transmitting the extended range PPDU for reception by a plurality of wireless devices that includes the second wireless device, the one or more processors are further arranged to:

in response to transmitting the extended range PPDU, simultaneously receive a non-extended range acknowledgement (ACK) from at least two of the plurality of wireless devices; and

transmit a second extended range PPDU for reception by the plurality of wireless devices.

20. A wireless device, comprising:

one or more wireless transceivers; and

one or more processors operably coupled to the one or more wireless transceivers, wherein the one or more processors are arranged to:

transmit a non-extended range Request to Send (RTS) frame;

receive, in response to the non-extended range RTS frame, an extended range Clear to Send (CTS) frame from a second wireless device; and

transmit a non-extended range physical layer protocol data unit (PPDU) for reception by the second wireless device.

21. The wireless device of claim **20**, wherein transmitting the non-extended range PPDU for reception by the second wireless device includes:

fragmenting the non-extended range PPDU into a plurality of non-extended range fragments;

transmitting a first non-extended range fragment of the plurality of non-extended range fragments;

receiving an extended range acknowledgement (ACK) from the second wireless device; and

transmitting a second non-extended range fragment of the plurality of non-extended range fragments.

22. The wireless device of claim **20**, wherein the one of more processors are further arranged to:

prior to transmitting the non-extended range PPDU, receive an extended range CTS frame from the second wireless device.

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